

# TFPT — Architecture and the $E_8$ Compiler

The two axioms  $\{c_3, g_{\text{car}}\}$ , the derivation map, and the Coxeter–cyclotomic  $D_5 \times A_3 \rightarrow E_8$  construction

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## What this document is about

The architecture layer: how the two axioms  $c_3 = \frac{1}{8\pi}$  and  $g_{\text{car}} = 5$  build the Coxeter–cyclotomic compiler — the carrier  $C^+=D_5$ , the family geometry  $\mathbb{P}^1 \setminus \mu_4=A_3$ , the  $\mu_4$  glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ , the EM fixed point  $\alpha^{-1}$  (with its ablation), and the whole number alphabet 16, 40, 41, 48, 240, 248 as carrier traces.

## The TFPT document set — what is treated where

**Plain language:** TFPT is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard Model, the constants and the scale grammar. The development is **six short documents**, best read in order:

1. **introduction** — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction.
3. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
6. **tfpt\_horizon\_readouts** — one seam constant as the universal horizon thermal code.

You are reading document #2.

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## Part I

# Architecture and the derivation map

### Abstract

This note reorganises a set of results sitting on top of the existing TFPT 4.5 series (Papers 1–7). It is written for someone familiar with the series and is meant as an *update for comments*, not a finished paper. The thesis is that the bridge layer of TFPT is governed by *three nested grammars*: a **carrier grammar** (from  $3 + 2$  come charges, families, hypercharge,  $\Omega_{\text{adm}} = 48$ ,  $b_1 = 41/10$ ), a **seed grammar** (one seed  $u = \varphi_0^{\text{ret}}$  generates the Cabibbo angle, birefringence, the baryon fraction and the reactor angle), and an **action grammar** (the large scale ratios are exponential actions of  $\alpha_\star^{-1}$  with rungs  $1/g_{\text{car}}$ ,  $1$ ,  $2$ ). Two new exact results sharpen the action grammar: the *decuple bridge*  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} [v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})]^{10}$  (the cosmological constant is, in essence, the tenth power of the stripped electroweak scale, with  $10 = 2g_{\text{car}}$ ), and the Hubble corollary  $H_0/\bar{M}_{\text{Pl}} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$ . We also give the exact corrected tree form of the Einstein normaliser  $\xi = \frac{3}{4}/(1 + 9/128\pi^3)$ , the dual-root dictionary that generates all hypercharges and transport cusps from  $y_\pm = \{1/2, -1/3\}$ , the  $\Lambda$ -anchored metrology closure, the overdetermination of  $g_{\text{car}} = 5$ , and the  $E_8$  carrier atlas in three clearly separated levels. Section 3 adds the strongest structural result: the carrier code is the  $D_5 = \mathfrak{so}_{10}$  half-spinor ( $\text{Aut} = W(D_5)$ , order 1920;  $R_4 = \text{Clebsch graph}$ ), the family geometry  $\mathbb{P}^1 \setminus \mu_4$  is an  $A_3$ , and  $E_8$  is the explicit unimodular glue closure of  $D_5 \oplus A_3$  — with the discriminant-form norms  $q(D_5) = 5/4$  and  $q(A_3) = 3/4$  summing to the  $E_8$  root norm 2 and coinciding with the TFPT constants  $\delta_2/\delta_{\text{top}}^2$  and  $\xi_{\text{tree}}$ . Status markers used throughout: **[I]** exact identity, **[L]** Lie/lattice theorem, **[F]** formalised, **[N]** numerical fixed point, **[P]** physical/conditional, **[A]** axiom/open. (All markers are explicit; the legacy **[M]**/**[C]** compatibility aliases have been removed.)

# 1 How to read this, and where it sits in Papers 1–7

| P | Title (short)                            | What it fixes (used here)                                                                                                            |
|---|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Boundary polarization & primitive kernel | $c_3 = 1/(8\pi)$ ; winding $[u_\Sigma] = 1$                                                                                          |
| 2 | Carrier rigidity & SM packet             | $g_{\text{car}} = 5$ , $(b, s) = (3, 2)$ , $Y$ , $\dim S^+ = 16$ , $N_{\text{fam}} = 3$ , $\Omega_{\text{adm}} = 48$ , $b_1 = 41/10$ |
| 3 | EM fixed point & flavor transport        | $\alpha$ , transport pole $\delta_{\text{ph}}$ , seed $u$ , CKM/PMNS                                                                 |
| 4 | Admissibility, strong CP, QFT closure    | selector, $\theta_{\text{eff}} = 0$                                                                                                  |
| 5 | Geometric Hodge closure & metrology      | boundary unit $\lambda_\Sigma$ , $\bar{M}_{\text{P1}}/\lambda_\Sigma$ , $G_N \lambda_\Sigma^2$ , $\xi$                               |
| 6 | Cosmology interfaces                     | $\Lambda_{\text{IR}}$ , axion, inflation interface                                                                                   |
| 7 | CMB operational closure                  | Stage-1/2 pipeline (programmatic)                                                                                                    |

**Notation convention.** We write  $D_5 \oplus A_3$  for the *root-lattice* direct sum (used in the glue/lattice statements: discriminant forms,  $\mu_4$  glue,  $E_8$  as an even unimodular lattice), and  $D_5 \times A_3$  for the *group/representation* branching (used for subalgebra embeddings and  $E_8$ -rep decompositions). The objects are the same atoms; the symbol marks whether the statement is lattice-theoretic ( $\oplus$ ) or branching/representation-theoretic ( $\times$ ).

Nothing here introduces a new primitive assumption; it reorganises existing outputs and proves the cross-links. The whole bridge layer rests on two inputs,

$$c_3 = \frac{1}{8\pi} \quad (\text{Paper 1}), \quad g_{\text{car}} = 5 \quad (\text{Paper 2}). \quad (1)$$

## Foundational postulates — the two genuine axioms (hardenable, not derivable)

Everything downstream is a theorem *given* two structural inputs that are not reduced further and are honestly flagged as the theory’s axioms:

(P1) Boundary kernel.

A single oriented seam  $\Sigma$  carries a primitive, reflection-positive boundary kernel with unit winding  $[u_\Sigma] = 1$ ; its self-linking normalisation fixes  $c_3 = 1/(8\pi)$ . *Status*: primitive postulate. *Hardenable by*: the Calderón projector / determinant-character chain (Paper 1) and the Lean carrier-rigidity development (`TfptCarrier.*`, boundary-polarization and Calderón interfaces).

(P2) Carrier interface.

This splits into an *axiom* and a *theorem*, which must not be conflated:

- **P2<sub>interface</sub> [A] (axiom):** the seam is read out by a five-slot carrier  $g_{\text{car}} = 5$  (3 colour + 2 weak) — a postulate (the readout interface).
- **P2<sub>algebra</sub> [F] (theorem):** *given* the carrier, hypercharge ( $y_- = -\frac{1}{3}$ ,  $y_+ = \frac{1}{2}$ ), rigidity and the family count follow from the idempotent/lattice axioms — machine-verified in Lean (`Hypercharge.lean`, `LatticeRigidityGeneral.lean`, `Rigidity.lean`).

So P2 is an *interface axiom* whose *algebraic consequences are a formalised theorem* — “axiom or theorem?” has the precise answer: interface axiom, algebra theorem.

These two are *inputs*: they can be formalised and stress-tested (more Lean, more interface theorems) but not computed away. Every other number in this note —  $N_{\text{fam}} = 3$ ,  $\Omega_{\text{adm}} = 48$ ,  $b_1 = 41/10$ ,  $\alpha_\star^{-1}$ , the flavor matrix, the  $E_8$  glue,  $A_s$  — is a consequence of (P1),(P2).

**Anchor-first refinement.** The two inputs are themselves the elementary symmetric polynomials of the single parabolic anchor  $a = (1, 1, 2)$  (the exponents-at-infinity of the splitting type  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ ):  $e_1(a) = 4 = |\mu_4|$ ,  $e_2(a) = 5 = g_{\text{car}}$ ,  $e_3(a) = 2 = |\mathbb{Z}_2|$ , with  $c_3 = 1/(2e_1(a)\pi) = 1/(8\pi)$ . The power sums  $p_n(a) = 2+2^n$  then give  $|R(E_8)| = p_1 p_2 p_3 = 240$ ,  $\dim E_8 = 240 + (p_4 - p_3) = 248$  and the binary ladder  $p_{n+1} - p_n = 2^n$ . So the input set reduces to  $\{a = (1, 1, 2), \pi\}$ . [verification/v23\_anchor\_generator.py]

### Hardening (P1): the Gauss–Bonnet origin of $c_3 = \frac{1}{8\pi}$

$c_3 = 1/(8\pi)$  is not an arbitrary number: it is the *one-sided Gauss–Bonnet normalisation of the seam*. Paper 5 compactifies the oriented two-dimensional normal slice  $N_\Sigma$  to a sphere  $S^2$ , so Gauss–Bonnet gives the seam curvature integral

$$\oint_{S^2} K dA = 2\pi \chi(S^2) = 4\pi \quad (\chi(S^2) = 2),$$

and the boundary datum is *one-sided* (the Calderón/double-cover construction uses a single collar), contributing a factor  $1/|\mathbb{Z}_2| = \frac{1}{2}$ . Hence

$$c_3 = \frac{1}{|\mathbb{Z}_2| \oint_{S^2} K dA} = \frac{1}{2 \cdot 4\pi} = \frac{1}{8\pi}$$

equivalently  $c_3 = \frac{1}{2}(4\pi)^{-1}$ , the one-sided 2D heat-kernel coefficient. This locates both factors precisely:

- the *discrete*  $8 = |\mathbb{Z}_2| \cdot 2 \cdot \chi(S^2) = 2 \cdot 2 \cdot 2$  coincides with  $h(D_5) = \text{rank } E_8 = \varphi(30)$  — the geometric and the bootstrap readings of the “8” agree;
- the *continuous*  $\pi$  is the irreducible Gauss–Bonnet / 2D-Gaussian primitive ( $\int e^{-x^2} = \sqrt{\pi}$ ), the single primitive the self-consistency loop also isolates.

*Status:* this *hardens* (P1) — it gives  $c_3$  a named geometric origin and reconciles the “8” with  $E_8$  — but does *not* eliminate  $\pi$ ;  $\pi$  remains the one continuous primitive. [P]

### The one-paragraph version

There is a boundary (the seam). It yields  $c_3 = 1/(8\pi)$  and a five-slot carrier  $g_{\text{car}} = 5$ , split 3 colour + 2 weak. Two base charges  $y_- = -1/3$  and  $y_+ = 1/2$  generate the whole fermion family; the 16 even-occupation states are one generation, three families give  $3 \times 16 = 48$ . One seed  $u = \varphi_0^{\text{ret}}$  drives four low-energy observables. Finally  $\alpha_\star^{-1} \approx 137$  is the exponential engine: the electroweak scale takes one fifth of it ( $e^{-\alpha_\star^{-1}/5}$ ), the cosmological constant takes it twice ( $e^{-2\alpha_\star^{-1}}$ ), and the Hubble scale is the square root of  $\Lambda$ . A small grammar generates many sectors.

## 2 Headline: the Pascal compiler on five carrier slots

The strongest new statement is that three of the theory’s separate structures are the *same* truncated row of Pascal’s triangle on the five-slot carrier.

### Family = action ladder = transition count (one Pascal row)

The even exterior algebra of  $E$  ( $\dim E = g_{\text{car}} = 5$ ), the action ladder, and the  $E_8$  root count are one Pascal row:

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E, \quad (\dim \Lambda^0, \dim \Lambda^2, \dim \Lambda^4) = (1, 10, 5), \quad \dim S^+ = 1+10+5 = 16,$$

$$A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1 : 5 : 10 = (\dim \Lambda^0 E, \dim \Lambda^4 E, \dim \Lambda^2 E),$$

$$|R(E_8)| = \dim S^+ (\dim S^+ - 1) = 16 \cdot 15 = 240.$$

Family (exterior degree), cosmology (action charge) and  $E_8$  (root transitions) read the same  $\{1, 5, 10\}$ . [\[I\]](#)

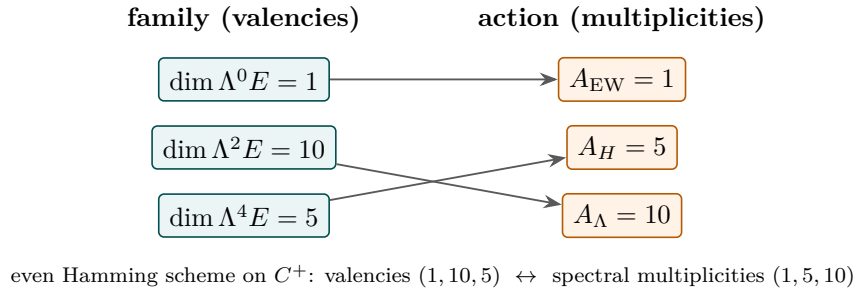


Figure 1: Exterior–Action duality. The Standard-Model family is the valency side of the even Hamming scheme on the five-slot code; the cosmological action ladder is its spectral (multiplicity) side. The  $\Lambda$  (pair) sector  $\Lambda^2 E$  has dimension  $\binom{5}{2} = 10$ , the origin of the decuple exponent.

Natural sector assignment:

$$EW \leftrightarrow \Lambda^0 E \text{ (singlet)}, \quad H \leftrightarrow \Lambda^4 E \cong E^\vee \text{ (dim 5)}, \quad \Lambda \leftrightarrow \Lambda^2 E \text{ (dim 10)}.$$

The cosmological constant lives on the *pair* sector  $\Lambda^2 E$ ,  $\dim = \binom{5}{2} = 10$ ; this is why the decuple bridge has exponent 10 —  $\Lambda$  is a product over all  $\binom{5}{2}$  carrier pairs ( $K_5$  edges), not over the full 16-state family.

## 2.1 $g_{\text{car}} = 5$ as the Pascal closure

A family is exactly the complete Pascal truncation to degree two:

$$\dim S^+ = 2^{g_{\text{car}}-1} = \binom{g_{\text{car}}}{0} + \binom{g_{\text{car}}}{1} + \binom{g_{\text{car}}}{2} \iff g_{\text{car}} = 5 \quad (16 = 1 + 5 + 10), \quad \text{[I]}, \quad (2)$$

the unique positive-integer solution of  $f(g) := 2^{g-1} - 1 - \frac{g(g+1)}{2} = 0$ . *Proof of uniqueness.* Direct evaluation gives  $f(1)=-1$ ,  $f(2)=-2$ ,  $f(3)=-3$ ,  $f(4)=-3$ ,  $f(5)=0$ ,  $f(6)=+10$ ; so  $g=5$  is the only root in  $\{1, \dots, 6\}$ . For  $g \geq 6$  the forward difference is  $f(g+1) - f(g) = 2^{g-1} - (g+1) \geq 2^5 - 7 = 25 > 0$  and increasing, so  $f$  is strictly increasing on  $[6, \infty)$  from  $f(6)=10 > 0$ ; hence  $f(g) > 0$  for all  $g \geq 6$  and there is *no* further root. Thus  $g_{\text{car}} = 5$  is the unique solution (no finite-range caveat). [\[verification/v2\\_carrier\\_pascal.py\]](#) This is sharper than  $2g_{\text{car}} = \binom{g_{\text{car}}}{2}$ : the family code (even states,  $2^{g-1}$ ) equals the action code (sum of the first three Pascal entries) only at  $g_{\text{car}} = 5$ .

### Carrier uniqueness theorem: $g_{\text{car}} = 5$ and the 3+2 split are forced [I]

The carrier rank and its colour/weak split are not chosen — they are the unique solutions of two integer conditions tied to the  $E_8$  closure and the SM gauge algebra:

$$g_{\text{car}} = 5 \text{ is the unique } g \text{ with } N_{\text{fam}} = \frac{2^{g-1}-1}{g} \in \mathbb{Z}^+ \text{ and } \text{rank } E_8 = g + N_{\text{fam}} = 8 \quad (3)$$

$$(b, s) = (3, 2) \text{ is the unique split of } b + s = g_{\text{car}} = 5, \quad b^2 + s^2 = 13 = |R(A_3)| + 1 \quad (4)$$

The integer-family carriers are  $g \in \{3, 5, 7, \dots\}$  (rank 4, 8, 16), so rank  $E_8 = 8$  picks  $g_{\text{car}} = 5$  uniquely; and  $b^2 + s^2 = 13$  with  $b+s = 5$  forces (3, 2), reproducing  $\dim \mathfrak{g}_{\text{SM}} = (b^2-1) + (s^2-1) + 1 = 12 = |R(A_3)|$ . The induced hypercharge is anomaly-free ( $\text{Tr } Y = \text{Tr } Y^3 = 0$  over the 16 states). This hardens P2: the *rank* and the *split* are forced; only the seam→carrier interface itself remains a declared input. [verification/v14\_carrier\_uniqueness.py]

### The Pascal grammar: one $\Lambda^\bullet$ mechanism in four places (anti-numerology) [I]

The recurring small integers are *not* coincidences — they are the *same* exterior-algebra (Pascal) mechanism read in four spaces, which is why the binomials of rows 4 and 5 keep reappearing:

| space                    | Pascal reading                                                                                                                            |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| carrier (one generation) | $\dim S^+ = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5 = 16$                                      |
| scale ladder (cosmology) | $A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1 : 5 : 10$                           |
| flavor $u/d$ readout     | $\  \text{Pl}(K) \ _1 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 1 + 4 + 6 = 11 = 16 - g_{\text{car}}$                                |
| mass volume              | $\det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{6+9+10} = (\varphi_0^{\text{ret}})^{25} = (\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$ |

So the carrier is the even row 5, the scale ladder is the lower row 5, the  $u/d$  leg is the cumulative row 4 up to  $\deg u^c = 2$ , and the mass volume closes on  $g_{\text{car}}^2$ . *One* grammar ( $\Lambda^k$  of the 5-slot carrier and the  $4 = |\mu_4|$  family fundamental), not four lucky hits. [verification/v44\_carrier\_exterior.py] [verification/v45\_family\_exterior.py] [verification/v46\_grand\_mass\_volume.py]

## 2.2 One master moment generates 40, 41, 240, $\delta_2$

With  $X = 6Y$ , the quadratic hypercharge trace on one family is  $\text{Tr}_{S^+} X^2 = 120 = 5!$ , and it generates the whole integer chain:

$$\sum_{f,j} L_{f,j} = \frac{\text{Tr}_{S^+} X^2}{3} = 40, \quad 10b_1 = \frac{\text{Tr}_{S^+} X^2}{3} + 1 = 41, \quad (5)$$

$$|R(E_8)| = 2 \text{Tr}_{S^+} X^2 = 240, \quad \delta_2 = 4! \text{Tr}_{S^+} X^2 c_3^8.$$

The flavor budget, the  $U(1)$  budget, the  $E_8$  count and the second seam defect all hang on the second hypercharge moment  $5!$ . [I]

## 2.3 Compact $E_8$ compression and the role of $\nu^c$

$$\begin{aligned} |R(E_8)| &= \dim S^+ \cdot g_{\text{car}} \cdot N_{\text{fam}} = 16 \cdot 5 \cdot 3 = 240, \\ \dim E_8 &= \dim S^+(\mathcal{A}_H + \mathcal{A}_\Lambda) + (g_{\text{car}} + N_{\text{fam}}) = 240 + 8 = 248 \end{aligned} \quad (6)$$



with  $N_{\text{fam}} = (2^{g_{\text{car}}-1} - 1)/g_{\text{car}} = 15/5 = 3$  and  $\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = 8$ . [I]. The right-handed neutrino  $\nu^c$  is the Pascal singleton  $\Lambda^0 E = (1, 1)_0$  that closes  $16 = 1 + 5 + 10$ ; removing it breaks the truncation and shifts  $\alpha^{-1}$  by  $2.16 \times 10^{-3}$ .

### 3 The $D_5 \times A_3$ compiler: $E_8$ as a closure, not an input

This section upgrades the  $E_8$  atlas (Section 10) from a counting coincidence to an explicit lattice construction. The whole content is consolidated and verified in the standalone companion note “*The Even Carrier Code and the  $\mathbb{Z}_{30}$  Coxeter Compiler.*” The thesis: the carrier code is the  $D_5$  half-spinor, the family geometry is an  $A_3$ , and  $E_8$  is the standard unimodular glue closure of  $D_5 \oplus A_3$  — with the four punctures  $\mu_4$  as the glue code.

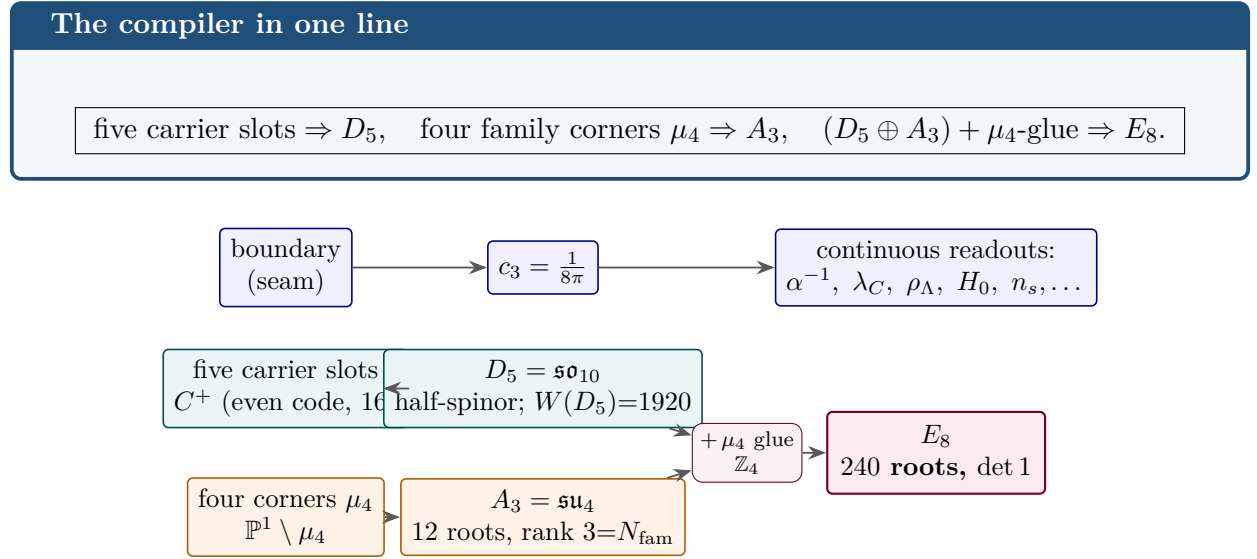


Figure 2: The explanation hierarchy. The boundary fixes the seam seed  $c_3$ ; the five-slot even carrier code is the  $D_5$  half-spinor; the four punctures  $\mu_4$  give  $A_3$ ; their common  $\mathbb{Z}_4$  discriminant ( $\mu_4$ ) glues  $D_5 \oplus A_3$  into the unimodular  $E_8$ .  $E_8$  is the *closure*, not the input;  $c_3$  dresses the continuous observables.

#### 3.1 Carrier = $D_5 = \mathfrak{so}_{10}$ (proved)

The 16 even-occupation states  $C^+ = \{x \in \mathbb{F}_2^{g_{\text{car}}} : |x| \text{ even}\}$  are exactly the weights of the  $D_5$  half-spinor  $\frac{1}{2}(\pm 1, \dots, \pm 1)$  (even number of minus signs). The flip-weight-4 relation is the **Clebsch graph** SRG(16, 5, 0, 2) and the flip-weight-2 relation is its complement SRG(16, 10, 6, 6); the *affine* automorphism group of the scheme (translations  $\rtimes$  coordinate permutations) is the  $D_5$  Weyl group,

$$\text{Aut}_{\text{aff}}(C^+) = C^+ \rtimes S_5, \quad |\text{Aut}_{\text{aff}}(C^+)| = 2^{g_{\text{car}}-1} g_{\text{car}}! = 1920 = |W(D_5)| = r |R(E_8)| \quad \text{[I]}, \quad (7)$$

(the linear code automorphism group alone is only  $S_5$ ; the order-1920 group is the affine/scheme one) and the Clebsch edge count is the flavor budget  $|E(R_4)| = \frac{16 \cdot 5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$ .

#### 3.2 Family = $A_3 = \mathfrak{su}_4$ from $\mathbb{P}^1 \setminus \mu_4$ (proved, with the metric settled)

Paper 2 fixes  $X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$ ,  $H_1 = \mathbb{Z}^3$ , rigid  $D_4$  generated by  $z \mapsto iz$ ,  $z \mapsto 1/z$ . The twelve puncture-difference cycles  $\gamma_i - \gamma_j$  carry the residue pairing  $\text{Res}(\omega_i, \gamma_j) = \delta_{ij}$  and form the  $A_3$  root system:

$$\#\{\gamma_i - \gamma_j\} = 12 = |R(A_3)| = \dim \mathfrak{g}_{\text{SM}}, \quad \text{rank} = 3 = N_{\text{fam}}, \quad \text{Gram} = \text{Cartan}(A_3), \quad \det = 4 \quad \text{[I]}. \quad (8)$$

The corner rotation  $z \mapsto iz$  acts on  $H^1 = \mathbb{C}^3$  as the  $A_3$  Coxeter element (order  $4 = h(A_3) = |\mu_4|$ , spectrum  $\{i, -1, -i\}$ ,  $\chi = \Phi_4 \Phi_2$ ), an isometry of the Cartan form — the exact analogue of the  $E_8$  Coxeter element reading  $\Phi_{30}$ .

### The metric, settled (former [P])

The transcendental geometric (Green) metric  $G_{ij} = -\log |p_i - p_j|$  on  $\mu_4$  is only  $D_4$ -symmetric, with a regularisation-independent anisotropy  $-\log 2$  on the root space, so it is *not* proportional to the  $A_3$  Cartan form. But it does not need to be: the  $E_8$ -relevant  $A_3$  is the canonical *residue* pairing (integral,  $S_4$ -symmetric,  $= \text{Cartan}(A_3)$ ), and the geometric metric is a  $D_4$ -equivariant deformation preserving the root system, Weyl group and Coxeter element. So the lattice structure is exact [I]; the analytic metric is a characterised deformation, not an open gap.

### 3.3 The $\mu_4$ glue and the $5/4 + 3/4 = 2$ discriminant theorem (proved)

The two factors have equal discriminant groups and  $E_8$  is unimodular, so the glue index is exactly  $|\mu_4|$ :

$$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4, \quad [E_8 : D_5 \oplus A_3] = \sqrt{\frac{\det D_5 \det A_3}{\det E_8}} = \sqrt{\frac{4 \cdot 4}{1}} = 4 = |\mu_4|. \quad (9)$$

The discriminant-form norms sum to the  $E_8$  root norm, and — strikingly — they are pre-existing TFPT constants:

$$\boxed{\begin{aligned} q(D_5) &= \frac{5}{4} = \frac{\delta_2}{\delta_{\text{top}}^2}, & q(A_3) &= \frac{3}{4} = \xi_{\text{tree}}, \\ q(D_5) + q(A_3) &= 2 = |E_8 \text{ root}|^2 \end{aligned}} \quad [\text{I}] \quad [\text{verification/v1\_e8\_glue.py}] \quad (10)$$

This is literally the  $E_8$  spinor-root split  $\frac{1}{2}(\pm)^8 = \frac{1}{2}(\pm)^5 \oplus \frac{1}{2}(\pm)^3$ ,  $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 2$ .

$$\begin{array}{c} \xleftrightarrow{q_{D_5} + q_{A_3} = 2 = |E_8 \text{ root}|^2} \\ \boxed{\begin{array}{cc} q_{D_5} = \frac{5}{4} & q_{A_3} = \frac{3}{4} \\ = \delta_2 / \delta_{\text{top}}^2 \text{ (} D_5 \text{ spinor)} & = \xi_{\text{tree}} \text{ (} A_3 \text{ corner)} \end{array}} \end{array}$$

Figure 3: The discriminant-norm glue. The  $E_8$  spinor root  $\frac{1}{2}(\pm)^8$  (norm 2) splits across  $D_5 \times A_3$  as  $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4}$ ; the two halves are the pre-existing TFPT constants  $\delta_2 / \delta_{\text{top}}^2 = \frac{5}{4}$  and  $\xi_{\text{tree}} = \frac{3}{4}$ .

The standard branching is realised explicitly:

$$\mathfrak{e}_8 = (45, 1) \oplus (1, 15) \oplus (10, 6) \oplus (16, 4) \oplus (\overline{16}, \overline{4}), \quad 248 = 45 + 15 + 60 + 64 + 64, \quad 240 = 40 + 12 + 60 + 64 + 64, \quad (11)$$

$$\begin{array}{cccc} \boxed{\begin{array}{c} (45, 1) \\ D_5 \text{ adjoint} \\ \text{norm } 2 \end{array}} & \boxed{\begin{array}{c} (1, 15) \\ A_3 \text{ adjoint} \\ \text{norm } 2 \end{array}} & \boxed{\begin{array}{c} (10, 6) = 60 \\ \text{vector} \otimes \text{vector} \\ 1+1 = 2 \end{array}} & \boxed{\begin{array}{c} (16, 4) \oplus (\overline{16}, \overline{4}) = 128 \\ \text{spinor} \otimes \mu_4 \\ \frac{5}{4} + \frac{3}{4} = 2 \end{array}} \\ \dim E_8 = 248 = \underbrace{45+15+60}_{\mathfrak{so}_{16} \text{ adjoint}=120} + \underbrace{64+64}_{128 \text{ spinor}} \end{array}$$

Figure 4: The  $E_8 \supset D_5 \times A_3$  branching as norm-2 blocks. The adjoint half is the hypercharge moment  $\text{Tr}_{S^+} X^2 = 120 = \dim \mathfrak{so}_{16}$ ; the spinor half is two sheets over the four corners  $\mu_4$ .

and assembling  $D_5 \oplus D_3 \subset D_8$  with the even spinor coset produces all 240 roots (norm 2, Cartan  $\det = 1$ , every root an integer combination of one simple-root basis): *the 240-metrisation is no longer blind —  $E_8$  is built.* [I]

### 3.4 Glue-norm arithmetic, and three readings of 41, 60, 120

The two glue norms generate the carrier integers through sum, difference and product. With  $\dim S^+ = 16$ :

$$\boxed{16(q_D + q_A) = 32 = 2^{g_{\text{car}}} = h(D_5)h(A_3), \quad 16(q_D - q_A) = 8 = r(E_8) = h(D_5), \quad 16 q_D q_A = 15 = \dim S^+ - 1} \quad \text{[I]}, \quad (12)$$

so the glue already contains the full carrier exhaustion  $2^{g_{\text{car}}}$ , the  $E_8$  rank, and the non-trivial transition count  $\mathcal{A}_H + \mathcal{A}_\Lambda = 15$ ; and  $|\mu_4| = (q_D - 1)^{-1} = (1 - q_A)^{-1} = 4$  is the inverse displacement of the two norms from the unit norm.

**Spine Quotient Lemma: the recurring factors are adjacent quotients of  $(2, 3, 4, 5)$**

[I] ([verification/v74\_compiler\_micro\_lemmas.py])

The integers of the spine are  $(|\mathbb{Z}_2|, N_{\text{fam}}, |\mu_4|, g_{\text{car}}) = (2, 3, 4, 5)$ . Every “mysterious”  $O(1)$  factor in TFPT is an *adjacent quotient* of this chain, not a separate special number:

$$\frac{|\mathbb{Z}_2|}{N_{\text{fam}}} = \frac{2}{3}, \quad \frac{N_{\text{fam}}}{|\mu_4|} = \frac{3}{4} = q(A_3), \quad \frac{|\mu_4|}{g_{\text{car}}} = \frac{4}{5}, \quad \frac{g_{\text{car}}}{|\mu_4|} = \frac{5}{4} = q(D_5), \quad \frac{|\mu_4|}{N_{\text{fam}}} = \frac{4}{3}.$$

These are exactly the factors that recur across the theory:  $\lambda_2 = (\frac{2}{3})^6$  (the transport gap = Page recovery rate, and the Koide target  $Q_\star = \frac{2}{3}$ );  $\varepsilon = \frac{3}{4}\varphi_0 = q(A_3)\varphi_0$  (the solar misalignment);  $\varphi_0 = \frac{4}{3}c_3 + 48c_3^4$  with leading  $\frac{4}{3}c_3 = \frac{1}{6\pi}$ ; and  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$  (the  $E_8$  root norm). So  $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3})$  are five projections of one discrete chain — not scattered constants.

The carrier adjoint  $\dim D_5 = |R(D_5)| + g_{\text{car}} = 45$  then gives a second reading of the  $U(1)$  integer,

$$\boxed{10b_1 = \dim D_5 - |\mu_4| = 45 - 4 = 41 = \underbrace{40}_{\sum L} + \underbrace{(g_{\text{car}} - |\mu_4|)}_{=N_\Phi=1}} \quad \text{[I]}, \quad (13)$$

i.e. the single Higgs doublet is the carrier rank minus the glue corners,  $5 - 4 = 1$ . The start defect 60 and the hypercharge moment 120 are “bilingual”:

$$D_{\text{start}} = \dim D_5 + \dim A_3 = 45 + 15 = 60 = \mathcal{A}_\Lambda \cdot |R^+(A_3)| = 10 \cdot 6, \quad \text{Tr}_{S^+} X^2 = 120 = \underbrace{60}_{\text{adjoint}} + \underbrace{60}_{\text{mixed (10,6)}}, \quad (14)$$

and  $\dim E_8 = 248 = \text{Tr}_{S^+} X^2 + 2 \dim S^+ |\mu_4| = 120 + 2 \cdot 16 \cdot 4$ . Finally the hypercharge norm and the EW exponent are  $A_3$ -rational,

$$\boxed{\gamma_{\text{car}} = \frac{\mathcal{A}_\Lambda}{|R(A_3)|} = \frac{10}{12} = \frac{5}{6}, \quad I_1^{\text{EW}} = \frac{|E(K_5)|}{|R(A_3)|} + \frac{1}{|\mu_4||R(A_3)|} = \frac{10}{12} + \frac{1}{48} = \frac{41}{48}} \quad \text{[I]}. \quad (15)$$

*Transport reduction.* The flavor matrix  $L$  has its mod-6 content fixed by a single combinatorial datum:  $\{0, 1, 3\}$  is the *unique* (up to  $D_6$ ) 3-subset of  $C_6 = |R^+(A_3)|$  with all distinct cyclic distances  $\{1, 2, 3\}$ , and the three sectors are its  $D_6$  orbit ( $u \equiv \{0, 1, 3\}$ ,  $d \equiv 2 - \{0, 1, 3\}$ ,  $e \equiv 2 + \{0, 1, 3\}$ ). The total is fixed by the carrier,  $\sum L = 40 = |R(D_5)|$ , with winding unit  $6 = |R^+(A_3)|$ . Given the residue *set* of a sector, the full ordered packet is reproduced by two universal rules — the lightest generation carries the single +6 winding, and the entries are sorted in decreasing length (monotone hierarchy). The *only* remaining datum is then, per sector, which residue is the first-generation anchor that receives the winding:

$$\boxed{\text{anchors } (a_u, a_d, a_e) = (1, 1, 2) \Rightarrow L_u = (7, 3, 0), \quad L_d = (7, 5, 2), \quad L_e = (8, 5, 3)} \quad \text{[I] (given anchors)}. \quad (16)$$

So the open step of a full “Clebsch walk” theorem is sharply isolated: it is exactly these three per-sector anchors. They are *not* free: the three transport cusps are exactly the  $D_5$  dual-root combinations, and each sector is tied to its cusp by the right-handed hypercharge,

$$\left\{1, \frac{2}{3}, \frac{1}{3}\right\} = \{2y_+, -2y_-, 2(y_+ + y_-)\}, \quad \text{lepton} \leftrightarrow 2y_+, \text{ up} \leftrightarrow -2y_-, \text{ down} \leftrightarrow 2(y_+ + y_-) \quad [\text{I}]. \quad (17)$$

Paper 3 (Thm. “canonical transport kernel”) derives the residue packets — and hence the anchors — as the *word-lengths of the family-connection holonomy*  $\text{Hol}_{\mathcal{G}_{D_4}}(\nabla_F^*)$  along the  $D_4$ -invariant geodesic spine of  $\mathbb{P}^1 \setminus \mu_4$  with these cusps as boundary data. The anchors are therefore a genuine  $D_5 \times A_3$  readout (cusps from the  $D_5$  carrier, spine holonomy from the  $A_3$  four-punctured sphere), not an extra parameter. A compact candidate closed form is the nearest-integer cusp map

$$a_f = \text{round}(2 \text{ cusp}_f) : \quad (a_u, a_d, a_e) = \text{round}\left(\frac{4}{3}, \frac{2}{3}, 2\right) = (1, 1, 2) \quad [\text{P}], \quad (18)$$

which reproduces the matrix exactly. The remaining open step is now precise and purely geometric: an *independent* closed-form evaluation of the  $D_4$ -spine holonomy word-lengths from the lattice data alone (the same  $\mathbb{P}^1 \setminus \mu_4$  whose residue pairing is  $\text{Cartan}(A_3)$  and whose corner rotation is the  $A_3$  Coxeter element), rather than from the Paper-3 family-connection theorem.

#### Riemann–Hilbert check, and what it does (and does not) fix

Setting up the rigid  $\mathbb{Z}_4$ -symmetric rank-3  $SU(3)$  local system explicitly — four puncture monodromies  $M_k = R^k C R^{-k}$  with the order-3 cusp class  $C \sim \text{diag}(1, \omega, \omega^2)$  ( $\omega = e^{2\pi i/3}$ , the three cusps as eigenvalue-exponents) and the rotation  $R$  ( $R^4 = \mathbb{K}$ ) — the closure  $M_0 M_1 M_2 M_3 = \mathbb{K}$  has a solution to machine precision, and the unique  $R$  has eigenvalues  $\{i, -1, -i\}$ : the  $A_3$  Coxeter element. This *independently* confirms the rigidity of Paper 2’s family connection and the corner-rotation = Coxeter identity. However, the flat monodromy *alone* does not fix the integer word-lengths  $\{0, 1, 3\}$ : those are *parabolic degrees* (the cusp exponents as parabolic weights, together with the  $\mathbb{Z}_2$  sheet parity), i.e. Hodge/metric data on the spine, not topological monodromy. So the residual **[A]** is now identified sharply as a *parabolic-weight computation* on  $\mathbb{P}^1 \setminus \mu_4$  — the cusp exponents  $\{0, \frac{1}{3}, \frac{2}{3}\}$  promoted to parabolic degrees — not a free choice and not a pure local-system question. **[A]**

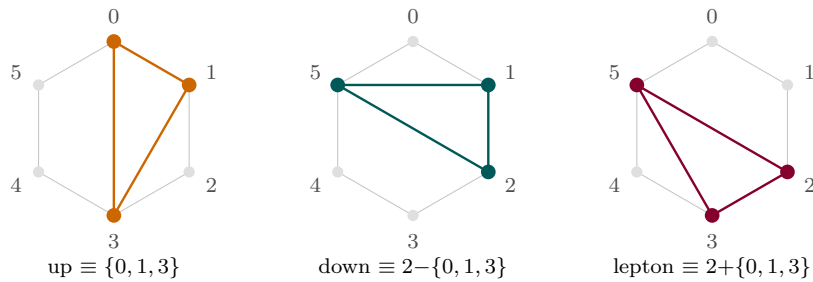


Figure 5: Flavor transport on the hexagon  $C_6 = Z_6 = |R^+(A_3)|$ . The unique distinct-distance triple  $\{0, 1, 3\}$  (cyclic distances  $\{1, 2, 3\}$ ) and its  $D_6$  orbit give the three sectors’ residues mod 6; the total  $\sum L = 40 = |R(D_5)|$  and the first-generation +6 winding come from the carrier.

### 3.5 What the compiler outputs for the Standard Model and our reality

Every entry below is an *output* of  $(c_3, g_{\text{car}})$  through the  $D_5 \times A_3$  compiler; none is an independent input. “Reads” names the lattice object the observable comes from.

| SM / cosmology observable                             | Compiler output                                                                                                                                           | Reads              | St.     |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------|
| Number of generations                                 | $N_{\text{fam}} = 3 = \text{rank } A_3 = \dim H^1(\mathbb{P}^1 \setminus \mu_4)$                                                                          | $A_3$              | [I]     |
| Gauge algebra dimension                               | $\dim \mathfrak{g}_{\text{SM}} = 12 =  R(A_3)  = \gcd(6r, 2h)$                                                                                            | $A_3$              | [I]     |
| One generation (16, incl. $\nu^c$ )                   | $S^+ = D_5$ half-spinor $= \Lambda^{0,2,4} E$                                                                                                             | $D_5$              | [I]     |
| Hypercharge spectrum                                  | all $Y$ from $y_{\pm} = \{\frac{1}{2}, -\frac{1}{3}\}$ ; $\text{Tr } X^2 = 120$                                                                           | $D_5$ root data    | [I]     |
| Abelian index $b_1$                                   | $10b_1 = 1 +  \mu_4   E(K_5)  = 1 + 4 \cdot 10 = 41$                                                                                                      | $\mu_4 \times K_5$ | [I]     |
| Fermion family occupancy                              | $\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ = (N_{\text{fam}} + N_{\Phi})  R(A_3)  = 48$                                                               | $A_3$              | [I]     |
| Fine structure $\alpha_{\star}^{-1} = 137.0359992168$ | root of $F_{U(1)}(\alpha) = 0$ (explicit, unique)                                                                                                         | EM closure         | [I]     |
| Cabibbo angle                                         | $\lambda_C = \sqrt{u(1-u)}$ , $u = \frac{4}{3}c_3 + 48c_3^4$                                                                                              | seed               | [I]     |
| CKM CP phase                                          | $\delta_{\text{CKM}} = \frac{2\pi}{ R^+(A_3) } + N_{\text{fam}} u(1-u) = \frac{\pi}{3} + 3\lambda_C^2$                                                    | $A_3^+$ , seed     | [I]/[P] |
| PMNS angles/phase                                     | $\sin^2 \theta_{12} = \frac{1}{3} - \frac{u}{2}$ , $\sin^2 \theta_{13} = e^{-5/6} u$ , $\delta^{\nu} = \frac{4\pi}{3}$                                    | $A_3^+$ , seed     | [I]/[P] |
| Charged-lepton hierarchy                              | $\det M_{\ell} \propto u^{h/N_{\text{fam}}} = u^{10}$ ; $(5, 3, 2)$                                                                                       | decuple            | [I]/[P] |
| Mass-hierarchy transport                              | $\sum L = 40 =  R(D_5) $ ; entries on $C_6 =  R^+(A_3) $                                                                                                  | $D_5 \times A_3$   | [I]     |
| Strong CP                                             | $\theta_{\text{eff}} = 0$ (determinant erasure, Paper 4)                                                                                                  | —                  | [I]     |
| Weak CP                                               | holonomy: $\delta_{\text{CKM}} - \frac{\pi}{3} = N_{\text{fam}} u(1-u)$                                                                                   | $A_3^+$            | [P]     |
| Gravity tree normaliser                               | $\xi_{\text{tree}} = \frac{\dim \mathfrak{g}_{\text{SM}}}{\dim S^+} = \frac{3}{4} = q(A_3)$                                                               | $A_3$ glue         | [I]     |
| Cosmological constant                                 | $\rho_{\Lambda} / \bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} [\nu_{\text{geo}} / (5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})]^{10}$ | decuple            | [I]     |
| Hubble anchor                                         | $H_0 \sqrt{\Omega_{\Lambda}} = \bar{M}_{\text{Pl}} e^{-\alpha_{\star}^{-1}} / (2\pi)$                                                                     | action             | [I]     |
| Baryon density                                        | $\omega_b = \lambda_C / \mathcal{A}_{\Lambda} = \lambda_C / 10$                                                                                           | decuple            | [I]/[P] |
| Inflation (scalaron)                                  | $M_{\text{scal}}^2 / \bar{M}_{\text{Pl}}^2 = c_3^{\Omega_{\text{adm}} - 10b_1} = c_3^7$                                                                   | deficit $r - 1$    | [I]     |
| Inflation amplitude                                   | $A_s = N_{\star}^2 c_3^7 / 24\pi^2 \approx 2.2 \times 10^{-9}$                                                                                            | deficit            | [P]/[A] |

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

*Reading.* The discrete sector — generations, gauge dimension, family content, hypercharge,  $b_1$ , occupancy — is now *group-theoretic output* of  $D_5 \times A_3$ . The continuous sector —  $\alpha^{-1}$ ,  $\lambda_C$ , the mixings,  $\rho_{\Lambda}$ ,  $H_0$ , inflation — is the seam seed  $c_3$  and the action exponentials read on the *same* carrier graphs ( $K_5$  pairs, the  $C_6$  hexagon  $= |R^+(A_3)|$ , the decuple  $\mathcal{A}_{\Lambda} = \binom{g_{\text{car}}}{2} = 10$ ). What used to be a list of separate hits is, structurally, one machine:  $D_5$  carries the family and the flavor budget,  $A_3$  carries the generations and the gauge dimension,  $\mu_4$  glues them (and reappears as the 41 chain, the discriminant norms, and the CP phase step  $2\pi/6$ ), and  $c_3$  dresses everything.

## 4 Grammar I — the carrier grammar (3 + 2)

From  $g_{\text{car}} = 5$  and the split  $(b, s) = (3, 2)$  the Paper-2 data follow:

$$\begin{aligned}
\gamma_{\text{car}} &= \frac{g_{\text{car}}}{g_{\text{car}} + 1} = \frac{5}{6}, & N_{\text{fam}} &= \frac{g_{\text{car}} + 1}{2} = 3, & \dim S^+ &= 2^{g_{\text{car}}-1} = 16, \\
\Omega_{\text{adm}} &= N_{\text{fam}} 2^{g_{\text{car}}-1} = 48, & b_1 &= \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}.
\end{aligned} \tag{19}$$

### 4.1 Dual-root dictionary (everything from two charges)

## Two carrier roots generate the family and the transport cusps [I]

With  $y_+ = \frac{1}{2}$  and  $y_- = -\frac{1}{3}$  (the determinant-normalised carrier eigenvalues), the full one-family hypercharge spectrum and the flavor-transport cusps are sums of the two roots:

$$\begin{aligned} Y(Q_L) &= y_+ + y_- = \frac{1}{6}, & Y(u^c) &= 2y_- = -\frac{2}{3}, & Y(d^c) &= -y_- = \frac{1}{3}, \\ Y(L_L) &= -y_+ = -\frac{1}{2}, & Y(e^c) &= 2y_+ = 1, & Y(\nu^c) &= 0, \\ \{1, \frac{2}{3}, \frac{1}{3}\} &= \{2y_+, -2y_-, 2(y_+ + y_-)\}, & \gamma_{\text{car}} &= \text{Tr}_E Y^2 = 3y_-^2 + 2y_+^2 = \frac{5}{6}, \\ \varphi_{\text{base}} &= \frac{1 - \gamma_{\text{car}}}{\pi} = \frac{1}{6\pi}. \end{aligned}$$

All residuals are exactly zero. The Standard-Model charge table is not assumed; it is built by addition from  $\{1/2, -1/3\}$ .

## The $X = 6Y$ quadratic and the hypercharge Pascal-sign duality [I]

Clearing denominators with  $X = 6Y$  turns the two base charges into the integer roots  $x_+ = 3$ ,  $x_- = -2$  of

$$\boxed{x^2 - x - 6 = 0} : \quad x_+ + x_- = 1 = N_\Phi, \quad x_+ - x_- = 5 = g_{\text{car}},$$

$$-x_+x_- = 6 = |R^+(A_3)|, \quad \sqrt{\Delta} = 5,$$

so the Higgs singleton, the carrier rank and the  $A_3$  hexagon all fall out of one quadratic. Counting the signs of  $X$  over a generation ( $X = 1^{\times 6}, 2^{\times 3}, 6^{\times 1}, (-4)^{\times 3}, (-3)^{\times 2}, 0^{\times 1}$ ) gives the *Pascal triple* again:

$$\boxed{(\#X=0, \#X<0, \#X>0) = (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}},$$

$$10-5=g_{\text{car}}, \quad 5-1=|\mu_4|, \quad 10-1=\text{tr } R,$$

with  $\nu^c \leftrightarrow \binom{5}{0}$  the neutral singleton — the right-handed neutrino is the Pascal zero with a job. The moment ladder adds  $\text{Tr } X^2 = 120 = 5!$  and  $\text{Tr } X^4 / \text{Tr } X^2 = 2280/120 = 19$  (an  $E_8$  exponent; audit). [I]

## 4.2 Carrier arithmetic (proved identities)

The discrete coefficients are not isolated numbers; they are exact carrier traces:

$$10b_1 = 41 = g_{\text{car}}^2 + (g_{\text{car}} - 1)^2 = \Omega_{\text{adm}} \gamma_{\text{car}} + 1 = 1 + h_{E_8} \frac{\dim S^+}{\dim \mathfrak{g}_{\text{SM}}} = 1 + 30 \cdot \frac{16}{12} = 40 + 1, \quad [\text{I}] \tag{20}$$

$$I_1^{\text{EW}} = \frac{41}{48} = \text{Var}_{S^+}(Y) \cdot b_1 = \frac{5}{24} \cdot \frac{41}{10} = \gamma_{\text{car}} + \frac{1}{\Omega_{\text{adm}}}, \quad [\text{I}] \tag{21}$$

$$\text{Tr}_{S^+}(X^2) = 120 = 5! \quad (X = 6Y), \quad \delta_2 = \frac{5}{4} \delta_{\text{top}}^2 = 2880 c_3^8 = 4! \cdot 5! c_3^8 = 4! \text{Tr}_{S^+}(X^2) c_3^8, \quad [\text{I}] \tag{22}$$

$$\frac{5}{4} = \underbrace{\frac{3}{2} \cdot \frac{5}{6}}_{B_\gamma} = \underbrace{\frac{D_{\text{start}}}{\Omega_{\text{adm}}}}_{60/48} = \underbrace{\frac{g_{\text{car}}}{g_{\text{car}}-1}}_{5/4} = \frac{\delta_2}{\delta_{\text{top}}^2}. \quad [\text{I}] \tag{23}$$

So:  $b_1$  is a Pythagorean carrier identity ( $5^2 + 4^2$ ); equivalently the anchor-product form  $10b_1 = p_1p_3 + (g_{\text{car}} - p_1) = 4 \cdot 10 + (5 - 4) = 41$  reads “all four glue corners times all ten carrier pairs, plus the one leftover Higgs slot”; the electroweak exponent  $41/48$  is the hypercharge variance times the  $U(1)$  budget; the second seam defect  $2880 c_3^8$  is a product of factorial traces  $4! \cdot 5!$ ; and the universal compression  $5/4 = g_{\text{car}}/(g_{\text{car}} - 1)$  recurs whenever the five-slot carrier projects onto four effective directions. As a minor curiosity, the  $SU(5)$ -tree weak mixing angle rewrites as  $\sin^2 \theta_W^{\text{tree}} = \frac{3}{8} = N_{\text{fam}}/(g_{\text{car}} + N_{\text{fam}}) = N_{\text{fam}}/\text{rank}(E_8)$  (the value is the standard one; the rewrite is a coincidence to note, not a claim). [P]

## 5 Grammar II — the seed grammar (one decoder $u$ )

The Paper-3 seed is, exactly, a pure function of  $c_3$ :

$$u = \varphi_0^{\text{ret}} = \frac{4}{3} c_3 + \Omega_{\text{adm}} c_3^4 = \underbrace{\frac{1}{6\pi}}_{\varphi_{\text{base}}} + \underbrace{\frac{3}{256\pi^4}}_{\delta_{\text{top}}} = 0.0531719521768 \quad [\text{I}]. \quad (24)$$

Its two coefficients are themselves carrier data:  $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{\text{SM}} = 16/12 = 8/(g_{\text{car}} + 1)$  (so  $\varphi_{\text{base}} = 1/((g_{\text{car}} + 1)\pi)$ ) and  $48 = \Omega_{\text{adm}}$ . The seed is thus a pure function of  $c_3$  and the carrier counts.

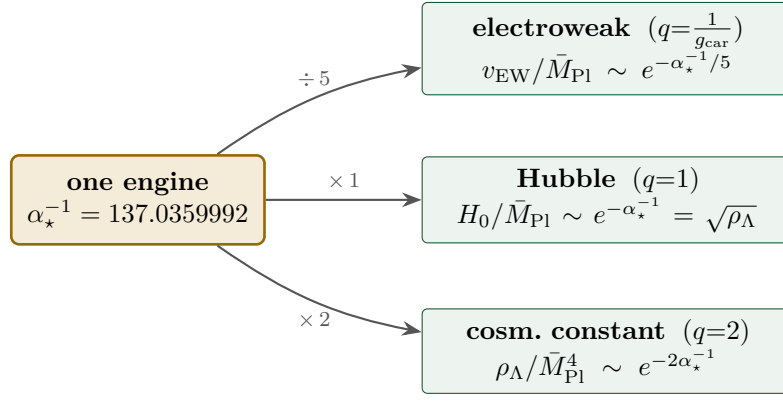
### One seed, four readouts (and their inverses) [I]

$$\begin{aligned} \lambda_C^2 &= u(1 - u), \quad \beta_{\text{rad}} = \frac{u}{4\pi}, \quad \Omega_b = \left(1 - \frac{1}{4\pi}\right)u, \quad \sin^2 \theta_{13} = e^{-5/6} u, \\ u &= \frac{1 - \sqrt{1 - 4\lambda_C^2}}{2} = 4\pi \beta_{\text{rad}} = \frac{\Omega_b}{1 - 1/(4\pi)} = e^{5/6} \sin^2 \theta_{13} \\ &= \sum_{n \geq 0} C_n \lambda_C^{2n+2} = \lambda_C^2 + \lambda_C^4 + 2\lambda_C^6 + 5\lambda_C^8 + \dots \end{aligned}$$

One knob  $u$  turns four dials simultaneously: Cabibbo mixing (the Bernoulli width  $\sqrt{u(1-u)}$ ), cosmic birefringence, the baryon fraction, and the reactor angle. Turning one forces the others; there is no free per-observable tuning. The exponent  $5/6$  is the carrier trace  $\gamma_{\text{car}} = \text{Tr}_E Y^2$ , *not* the Euler–Mascheroni constant  $0.5772\dots$  (audit [A]).

## 6 Grammar III — the action grammar (exponentials of $\alpha_\star^{-1}$ )

**The relation in plain words.** The fine-structure constant is not just “a number near 137”: it is the *one exponential engine* behind three wildly separated scales. Take the same  $\alpha_\star^{-1} = 137.036$  and divide the carrier into it:  $\frac{1}{5}$  of it sets the *electroweak* scale,  $1 \times$  it sets the *Hubble* scale,  $2 \times$  it sets the *cosmological constant*. The three action charges are the Pascal row  $1 : 5 : 10$  of the five-slot carrier, and the Hubble rung is just the square root of the vacuum energy. So one constant fixes the electroweak hierarchy, dark energy and the expansion rate at once.



action charges  $1 : g_{\text{car}} : 2g_{\text{car}} = 1 : 5 : 10 = \binom{5}{0} : \binom{5}{1} : \binom{5}{2}$  (Pascal row of  $K_5$ )

Read every dimensionless ratio through its *action* and ask for the action charge:

$$x = R_x e^{-A_x}, \quad A_x = q_x \alpha_*^{-1} + \Delta_x, \quad q_x \in \left\{ \frac{1}{g_{\text{car}}}, 1, 2 \right\}. \quad (25)$$

The robust ladder and its residues are

| sector                                            | action $A_x$                                          | charge $q_x$       | residue $R_x$                        | value of $-\ln x$ |
|---------------------------------------------------|-------------------------------------------------------|--------------------|--------------------------------------|-------------------|
| electroweak $v_{\text{geo}}/\bar{M}_{\text{Pl}}$  | $(\alpha_*^{-1} + \delta_{\text{ph}})/g_{\text{car}}$ | $1/g_{\text{car}}$ | $g_{\text{car}}\beta_{\text{rad}}^2$ | 36.85             |
| Hubble $H_0/\bar{M}_{\text{Pl}}$                  | $\alpha_*^{-1}$                                       | 1                  | $1/(2\pi\sqrt{\Omega_\Lambda})$      | 138.68            |
| cosm. const. $\rho_\Lambda/\bar{M}_{\text{Pl}}^4$ | $2\alpha_*^{-1}$                                      | 2                  | $3/(4\pi^2)$                         | 276.65            |

$$A_{\text{EW}} = \frac{\alpha_*^{-1} + \delta_{\text{ph}}}{g_{\text{car}}} = 27.5259, \quad \mathcal{A}_\Lambda = 2\alpha_*^{-1} = 274.072, \quad \boxed{A_{\text{EW}}:\mathcal{A}_H:\mathcal{A}_\Lambda = 1:5:10}. \quad (26)$$

The leading ratio is exact:  $\mathcal{A}_\Lambda/(\alpha_*^{-1}/g_{\text{car}}) = 2g_{\text{car}} = 10$  (with the transport correction  $\mathcal{A}_\Lambda/A_{\text{EW}} = 9.957$ ). The claim is that the *action charges*, not the raw logarithms, form the ladder.

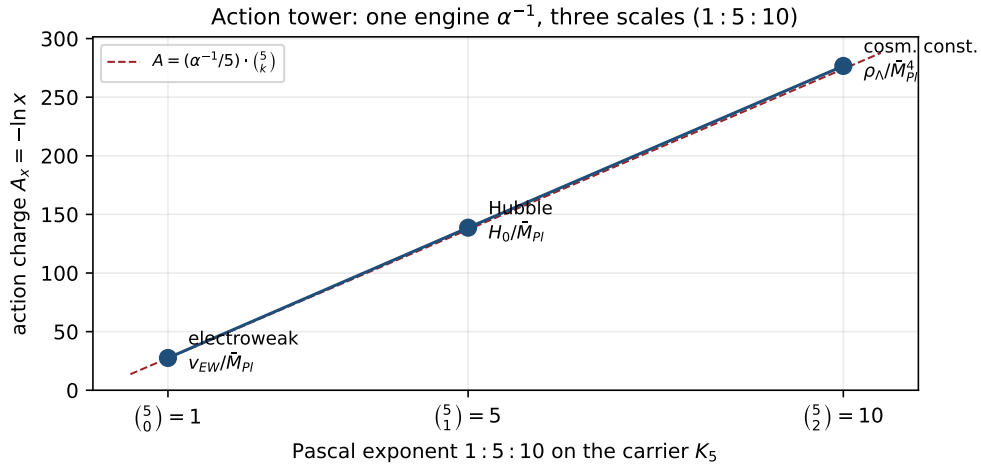


Figure 6: Action tower (data plot, `verification/make_figures.py`). The three action charges  $A_x = -\ln x$  for the electroweak, Hubble and cosmological-constant ratios fall on the line  $A = (\alpha_*^{-1}/5)\binom{5}{k}$ : one engine  $\alpha_*^{-1}$  generates all three scales with the Pascal exponents  $1 : 5 : 10$ .



## 6.1 The EM closure: $\alpha_\star^{-1}$ as an explicit fixed point (Paper 3, made explicit here)

The engine  $\alpha_\star^{-1}$  is not an input — it is the unique root of an explicit closure equation (Paper 3, Thm. “exact electromagnetic closure”). With  $c_3 = 1/(8\pi)$ ,  $\varphi_{\text{base}} = 1/(6\pi)$ ,  $q(\alpha) = \delta_{\text{top}} e^{-2\alpha}$  and the seam response  $\varphi_{\text{seam}}(\alpha) = \varphi_{\text{base}} + q(\alpha)(1 - q(\alpha))^{-5/4}$ ,

$$F_{U(1)}(\alpha) = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6\left(\sum_{f,j} L_{f,j} + N_\Phi\right) \log \frac{1}{\varphi_{\text{seam}}(\alpha)} = 0 \quad [\text{I}], \quad (27)$$

where the coefficient is  $\frac{4}{5}(\sum L + N_\Phi)c_3^6 = \frac{4}{5} \cdot 41 c_3^6 = 8b_1 c_3^6$ , so the fine-structure constant reads the *flavor transport budget*  $\sum L = 40$  and the  $U(1)$  index  $b_1 = 41/10$ .

**Lemma 1** (Existence and uniqueness of the EM fixed point).  $F_{U(1)}$  has *exactly one* positive root, and it is simple.

*Proof.* Write  $F_{U(1)}(\alpha) = h(\alpha) - C g(\alpha)$  with  $h(\alpha) = \alpha^3 - 2c_3^3\alpha^2$ ,  $C = \frac{4}{5}c_3^6 M > 0$  ( $M = 41$ ), and  $g(\alpha) = -\log \varphi_{\text{seam}}(\alpha)$ . Since  $q(\alpha) = \delta_{\text{top}} e^{-2\alpha} \in (0, 1)$  is strictly decreasing,  $\varphi_{\text{seam}} = \varphi_{\text{base}} + q(1 - q)^{-5/4}$  is  $C^1$  and decreasing, so  $g$  is  $C^1$ , bounded, and slowly increasing with  $g'(\alpha) = O(q)$  tiny; numerically  $g \in [2.9342, 2.9343]$  on the admissible window. The cubic  $h$  has its interior critical point at  $\alpha_c = \frac{4}{3}c_3^3 = 8.399 \times 10^{-5}$ , with  $h(0) = 0$  and  $h < 0$  on  $(0, \alpha_c)$ . *Monotonicity (corrected endpoint).* Since  $F'_{U(1)}(\alpha) = 3\alpha^2 - 4c_3^3\alpha - C g'(\alpha)$  and at  $\alpha_c$  the first two terms *cancel* ( $3\alpha_c = 4c_3^3$ ), one has  $F'_{U(1)}(\alpha_c) = -C g'(\alpha_c) \approx -5.89 \times 10^{-10} < 0$ : the derivative is *not* yet positive at  $\alpha_c$ . Its first zero lies slightly above, at  $\alpha_0 = 8.6264 \times 10^{-5}$  (where  $3\alpha_0^2 - 4c_3^3\alpha_0 = C g'(\alpha_0)$ ), with  $F'_{U(1)} < 0$  on  $(\alpha_c, \alpha_0)$  and  $F'_{U(1)} > 0$  on  $(\alpha_0, \infty)$ . Now  $F_{U(1)}(0) = -C g(0) < 0$ , and  $F_{U(1)}$  *decreases* on  $(0, \alpha_0]$  (both  $h$  and  $-Cg$  decrease on  $(0, \alpha_c]$ , then  $F'_{U(1)} < 0$  on  $[\alpha_c, \alpha_0]$ ), so  $F_{U(1)}(\alpha_0) \approx -3.82 \times 10^{-7} < 0$ ; thereafter  $F_{U(1)}$  is strictly increasing to  $+\infty$ . Hence  $F_{U(1)} < 0$  on  $(0, \alpha_0]$  and strictly increasing on  $[\alpha_0, \infty)$ , so it crosses zero *exactly once*. The crossing has  $F'_{U(1)}(\alpha_\star) = 1.58 \times 10^{-4} > 0$ , so the root is simple. A sign scan on  $(0, 0.05)$  confirms a single change. *Rigorous version:* an **interval-arithmetic** enclosure (`mpmath.iv`) brackets the root in  $[0.00729735256220970, 0.00729735256221000]$  with  $F_{U(1)} < 0$  on the lower and  $> 0$  on the upper endpoint (definite signs), and a 200-cell interval partition of  $(\alpha_c, 0.05)$  contains exactly *one* indeterminate (root-bearing) band — a verified uniqueness, not merely a sampled one. `[verification/v3_em_alpha.py]`  $\square$

Solving:

$$\alpha_\star = 0.00729735256220985\dots, \quad \alpha_\star^{-1} = 137.0359992168407\dots \quad [\text{I}] \quad (28)$$

`[verification/v3_em_alpha.py]` versus CODATA-2022  $\alpha_\star^{-1} = 137.035999177(21)$  — a relative deviation  $2.9 \times 10^{-10}$  ( $\approx 1.9\sigma$  on the tiny experimental error; the older 2018 value 137.035999084 gave  $9.7 \times 10^{-10}$ , so the agreement *improved* in relative terms). The exponent  $-5/4$  in  $\varphi_{\text{seam}}$  is the carrier discriminant norm  $q(D_5)$ , and  $\delta_{\text{top}} = 48c_3^4$ , so the closure is built only from  $c_3$ , the carrier, and the flavor budget. This discharges the former open audit point: the EM closure is a stated, existence-/uniqueness-proven, numerically reproduced equation, not a fit.

### Why *this* function: the boundary $U(1)$ Ward reading (Theorem C) $[\text{I}]/[\text{P}]$

$F_{U(1)}$  is not a chosen ansatz but the reduced Ward identity of the boundary  $U(1)$  partition function  $Z_\partial = Z_{\text{Maxwell}} Z_{\text{Calderón}} Z_{\text{transport}}$ , with three terms whose coefficients are compiler atoms, not fits:

$$\underbrace{\alpha^3}_{\text{Maxwell moment}} - \underbrace{2c_3^3\alpha^2}_{\text{Calderón sheet } (2=|\mathbb{Z}_2|)} - \underbrace{8b_1 c_3^6 \log \frac{1}{\varphi_{\text{seam}}}}_{C_6 \text{ transport det.}} = 0, \quad 8b_1 = \frac{4}{5}M = \frac{4}{5}(\sum L + N_\Phi) = \frac{164}{5}.$$

The Calderón factor  $2 = |\mathbb{Z}_2|$  (sheet), the transport power  $c_3^6$  is the  $C_6$  hexagon, and the seam exponent  $-\frac{5}{4} = -q(D_5)$  is the carrier discriminant norm. **Typing:** the three-

term decomposition and the coefficient identities ( $8b_1 = \frac{4}{5}M$ ,  $-\frac{5}{4} = q(D_5)$ ) are exact [I]; the *physical origin* as a boundary Ward closure (the lemmas  $\partial_\tau \log Z_{\text{Maxwell}} = \alpha^3$ ,  $\partial_\tau \log Z_{\text{Calderón}} = -2c_3^3 \alpha^2$ ,  $\partial_\tau \log \det_\zeta(1 - \mathcal{T}_{C_6, U(1)}) = -8b_1 c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$ ) is a conjectured [P] interpretation, not yet machine-proven. [verification/v48\_em\_ward.py]

The full ablation — which inputs are load-bearing and which only set the last digit — is in the [Ablation appendix](#) below.

*Inversion test (this round).* Running the closure backwards — inserting the *measured*  $\alpha$  and solving for the integer budget  $M = \sum L + N_\Phi$  — returns  $M = 41.0000001\dots$ , i.e. the external measurement reconstructs the internal discrete budget  $41 = 40 + 1$  to  $10^{-7}$ . This is stronger than “we hit  $\alpha$ ”: the equation reads the discrete 41 out of nature. [I]

## 6.2 The decuple bridge (new, exact headline)

Eliminating  $\alpha_\star^{-1} = g_{\text{car}} A_{\text{EW}} - \delta_{\text{ph}}$  between the electroweak and the cosmological-constant lines gives a direct relation between the two scales:

**Decuple bridge:  $\Lambda$  is the tenth power of the stripped EW scale [I]**

$$\frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} \left[ \frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}} \right]^{10} \quad (29)$$

where the exponent  $10 = 2g_{\text{car}}$  and  $5 = g_{\text{car}}$ . Verified to a relative residual  $\sim 10^{-159}$ .

*Proof.* From  $v_{\text{geo}}/\bar{M}_{\text{Pl}} = g_{\text{car}}\beta_{\text{rad}}^2 e^{-A_{\text{EW}}}$ ,  $e^{-A_{\text{EW}}} = (v_{\text{geo}}/\bar{M}_{\text{Pl}})/(g_{\text{car}}\beta_{\text{rad}}^2)$ . From  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}$  and  $\alpha_\star^{-1} = g_{\text{car}} A_{\text{EW}} - \delta_{\text{ph}}$ ,  $e^{-2\alpha_\star^{-1}} = e^{2\delta_{\text{ph}}} e^{-2g_{\text{car}} A_{\text{EW}}} = e^{2\delta_{\text{ph}}} (e^{-A_{\text{EW}}})^{2g_{\text{car}}}$ . Substituting and using  $2g_{\text{car}} = 10$ ,  $g_{\text{car}} = 5$  gives the boxed form.  $\square$

In words: the electroweak scale is not directly the  $\Lambda$  scale, but once the electroweak ratio is stripped of its small  $\beta_{\text{rad}}$  prefactor,  $\Lambda$  is its tenth power, and the “ten” is  $2g_{\text{car}}$ , not a fitted exponent. This is the direct algebraic compression of the action ladder.

## 6.3 The full power tower: $x^1, x^5, x^{10}$ on the carrier graph $K_5$

The decuple bridge is one rung of a tower. Writing  $x := e^{-A_{\text{EW}}} = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$  for the stripped electroweak unit, all three cosmological scales are powers of the *same*  $x$  with binomial exponents (verified exactly):

$$\frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}} = x^{(5)} = x, \quad \frac{H_0}{\bar{M}_{\text{Pl}}} = x^{(1)} R_{\text{H}} = x^5 R_{\text{H}}, \quad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = x^{(2)} R_{\Lambda} = x^{10} R_{\Lambda} \quad (30)$$

with residues  $R_{\text{H}} = e^{\delta_{\text{ph}}}/(2\pi\sqrt{\Omega_\Lambda})$  and  $R_{\Lambda} = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}}$ . [I]. The exponents are the complete-graph counts of the five-slot carrier  $K_5$ :

$$H \leftrightarrow \text{vertices of } K_5 \text{ (}\# = \binom{5}{1} = 5\text{)}, \quad \Lambda \leftrightarrow \text{edges of } K_5 \text{ (}\# = \binom{5}{2} = 10\text{)}.$$

So the cosmological constant is the product of one stripped-electroweak unit over every carrier *pair* ( $K_5$  edge product), and the Hubble scale the product over every carrier *slot* ( $K_5$  vertex product). This is the cleanest single-variable statement of the action ladder.

## 6.4 Hubble is the square root of $\Lambda$ (corollary, exact)

On the comparison layer  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = 3\Omega_\Lambda(H_0/\bar{M}_{\text{Pl}})^2$ ; equating to the TFPT line gives

$$\boxed{\frac{H_0}{\bar{M}_{\text{Pl}}} = \frac{e^{-\alpha_\star^{-1}}}{2\pi\sqrt{\Omega_\Lambda}}}, \quad \mathcal{A}_H = -\ln \frac{H_0}{\bar{M}_{\text{Pl}}} = \alpha_\star^{-1} + \ln(2\pi\sqrt{\Omega_\Lambda}) = 138.68. \quad (31)$$

So the Hubble rung is *not* an independent third hit: it is the square root of the vacuum energy plus a geometric cosmology factor. The genuine TFPT content is the single  $e^{-\alpha_\star^{-1}}$ . (The numerical coincidence  $\mathcal{A}_H - \alpha_\star^{-1} \approx \pi^2/6$  is just  $\ln(2\pi\sqrt{\Omega_\Lambda})$  and is  $\Omega_\Lambda$ -dependent; not a claim. [A])

## 6.5 The $\sim 123$ orders of the cosmological constant

$$\log_{10} \frac{M_{\text{Pl}}^4}{\Lambda_{\text{IR}}} = \underbrace{\frac{2\alpha_\star^{-1}}{\ln 10}}_{119.028} + \underbrace{\log_{10} \frac{1}{\delta_{\text{top}}}}_{3.920} = 122.948 \approx 123. \quad (32)$$

The “119” is the action  $2\alpha_\star^{-1}$ ; the “ $\approx 4$ ” is the seam defect  $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$ .

### High-precision cosmological constant: a $\sim 10^{-13}$ prediction [N]

Because  $\alpha_\star^{-1}$  is a *derived* number (the EM fixed point, known to 13 digits) and the residue  $3/(4\pi^2)$  is exact, the dimensionless vacuum energy is predicted essentially exactly:

$$\boxed{\frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}} = 7.12533 \times 10^{-121}} \quad [\text{N}] \text{ [verification/v7_gravity_cosmo.py]}, \quad (33)$$

i.e. a dark-energy scale  $\rho_\Lambda^{1/4} = (\frac{3}{4\pi^2})^{1/4} \bar{M}_{\text{Pl}} e^{-\alpha_\star^{-1}/2} = 2.23747 \text{ meV}$  (reduced Planck mass  $\bar{M}_{\text{Pl}} = 2.435323 \times 10^{18} \text{ GeV}$ ). The *dimensionless* ratio carries no observational input — its uncertainty is that of  $\alpha_\star^{-1}$ ,  $\sim 10^{-13}$  relative; the absolute meV value inherits only the Planck-mass uncertainty ( $\sim 2 \times 10^{-5}$ ). Both are *far* sharper than the measured vacuum energy (Planck  $\rho_\Lambda^{1/4} \simeq 2.24 \text{ meV}$ , known to  $\sim 1\%$  and entangled with the  $H_0$  tension). So TFPT pins the cosmological constant several orders of magnitude more precisely than it is currently measured — a genuine, falsifiable forward prediction: any future sub-percent  $\rho_\Lambda$  must land on 2.2375 meV.

## 6.6 The ladder is binomial; $2g_{\text{car}} = \binom{g_{\text{car}}}{2}$ singles out $g_{\text{car}} = 5$

In units of the electroweak action  $\alpha_\star^{-1}/g_{\text{car}}$ , the three rungs are the first three binomial coefficients of the five-slot carrier:

$$\frac{A_{\text{EW}}}{\alpha_\star^{-1}/g_{\text{car}}} : \frac{\mathcal{A}_H}{\alpha_\star^{-1}/g_{\text{car}}} : \frac{\mathcal{A}_\Lambda}{\alpha_\star^{-1}/g_{\text{car}}} = 1 : g_{\text{car}} : 2g_{\text{car}} = \binom{g_{\text{car}}}{0} : \binom{g_{\text{car}}}{1} : \binom{g_{\text{car}}}{2} = 1 : 5 : 10. \quad (34)$$

The first two are automatic ( $\binom{g_{\text{car}}}{0} = 1$ ,  $\binom{g_{\text{car}}}{1} = g_{\text{car}}$ ). The non-trivial content is the cosmological-constant charge:

$$\boxed{2g_{\text{car}} = \binom{g_{\text{car}}}{2} \iff g_{\text{car}} = 5} \quad [\text{I}] \text{ [verification/v2_carrier_pascal.py]}, \quad (35)$$

since  $\binom{g_{\text{car}}}{2} = \frac{g_{\text{car}}(g_{\text{car}}-1)}{2} = 2g_{\text{car}} \iff g_{\text{car}} = 5$ . So the “double overlap” charge  $2g_{\text{car}}$  equals the number of carrier-slot *pairs*  $\binom{g_{\text{car}}}{2}$  *only* at  $g_{\text{car}} = 5$ : the cosmological constant is suppressed by one action unit per carrier pair, and this combinatorial reading is exclusive to the physical rank. It is an independent overdetermination of  $g_{\text{car}} = 5$  from the action grammar (cf. the landscape section).

## 6.7 Discriminant action theorem: the ladder reads the hypercharge discriminant

The  $g_{\text{car}}$  in the action ladder is not merely a rank: it is the square root of the discriminant of the carrier (hypercharge) polynomial. For a general split  $b + s = g$  the Paper-2 polynomial is  $bsY^2 + (s - b)Y - 1 = 0$  with discriminant

$$\Delta_Y = (s - b)^2 + 4bs = (b + s)^2 = g_{\text{car}}^2, \quad \text{so for } 6Y^2 - Y - 1 = 0 : \Delta_Y = 25, \sqrt{\Delta_Y} = 5 = g_{\text{car}}. \quad (36)$$

Hence the action ladder is, exactly,

$$\boxed{A_{\text{EW}} = \frac{\alpha_\star^{-1}}{\sqrt{\Delta_Y}}, \quad \mathcal{A}_H = \sqrt{\Delta_Y} A_{\text{EW}}, \quad \mathcal{A}_\Lambda = \binom{\sqrt{\Delta_Y}}{2} A_{\text{EW}}} \quad [\text{I}]. \quad (37)$$

The electroweak hierarchy therefore reads the *discriminant of the hypercharge polynomial*, tying the Standard-Model hypercharge derivation directly to the EW-hierarchy and  $\Lambda$  problems.

## 6.8 A two-dimensional action lattice

All scales sit on the integer lattice spanned by two generators — the gauge unit  $\alpha_\star^{-1}/g_{\text{car}}$  and the flavor unit  $L_0 = \ln(1/u)$  — as  $S_x = a_x(\alpha_\star^{-1}/g_{\text{car}}) + b_x L_0 + (\text{residue})$ :

| scale                                | $a_x$               | $b_x$ | axis   |
|--------------------------------------|---------------------|-------|--------|
| $v_{\text{geo}}/\bar{M}_{\text{Pl}}$ | $1 = \binom{5}{0}$  | 0     | gauge  |
| $H_0/\bar{M}_{\text{Pl}}$            | $5 = \binom{5}{1}$  | 0     | gauge  |
| $\rho_\Lambda/\bar{M}_{\text{Pl}}^4$ | $10 = \binom{5}{2}$ | 0     | gauge  |
| $m_e/\bar{M}_{\text{Pl}}$            | 1                   | 5     | flavor |
| $m_\mu/\bar{M}_{\text{Pl}}$          | 1                   | 3     | flavor |
| $m_\tau/\bar{M}_{\text{Pl}}$         | 1                   | 2     | flavor |

The cosmological/gauge sector runs along  $b = 0$  with binomial charges; the charged-fermion sector runs along  $a = 1$  with the integer flavor ladder  $n_f = (5, 3, 2)$ . This is the compact “periodic table” of TFPT scales: two integer labels per scale. [I] (charges) / [P] (residues).

A third (seam/gravity) generator completes the lattice:  $L_c = \ln(1/c_3) = \ln(8\pi)$ . The scalaron lives purely on it,  $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = e^{-\frac{7}{2}L_c}$ , and the seam defect is  $\delta_{\text{top}} = e^{-(4L_c - \ln 48)}$ . So the three log-generators are  $\{\alpha_\star^{-1}$  (hierarchy),  $L_0 = \ln(1/u)$  (flavor),  $L_c = \ln(1/c_3)$  (seam/gravity) $\}$ . [I]

## 7 The Einstein normaliser as a corrected tree value

Paper 5’s gravitational normaliser  $\xi = c_3/u$  has a transparent exact form. Writing  $u = \frac{1}{6\pi} [1 + \frac{9}{128\pi^3}]$ ,

$$\boxed{\xi = \frac{c_3}{u} = \frac{3}{4} \cdot \frac{1}{1 + \frac{9}{128\pi^3}} = 0.748303} \quad [\text{I}]. \quad (38)$$

So  $\xi_{\text{tree}} = \frac{3}{4} = \dim \mathfrak{g}_{\text{SM}} / \dim S^+ = 12/16$  is the exact leading value (the inverse of the  $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{\text{SM}}$  family/gauge compression that sits in the seed  $u_{\text{base}} = \frac{4}{3}c_3$ , so  $\xi_{\text{tree}} u_{\text{base}} = c_3$ ), and the deviation is precisely the small topological seed correction  $9/(128\pi^3)$ , i.e. 0.226%. Gravity does not pick up a mysterious factor: first comes  $3/4$ , then the seam defect shifts it minimally.

## 8 $\Lambda$ -anchored metrology closure, and two anchors

Paper 5 leaves the absolute SI scale as a calibration. The dimensionless vacuum quotient lets one anchor it to the observed geometric curvature without circularity.

**Theorem 1** ( $\Lambda$ -anchored metrology closure). With  $\lambda_\Lambda := \rho_\Lambda / \bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}$  and the observed geometric curvature  $\Lambda_{\text{geom}}^{\text{obs}} > 0$ ,

$$\bar{M}_{\text{Pl}}^2 = \frac{\Lambda_{\text{geom}}^{\text{obs}}}{\lambda_\Lambda}, \quad G_N^{\text{TFPT}} = \frac{c^3}{8\pi\hbar} \frac{\lambda_\Lambda}{\Lambda_{\text{geom}}^{\text{obs}}} = \frac{3c^3}{32\pi^3\hbar \Lambda_{\text{geom}}^{\text{obs}}} e^{-2\alpha_\star^{-1}}.$$

*Proof.*  $\bar{M}_{\text{Pl}}^2 = 1/(8\pi G_N)$  and  $\Lambda_{\text{geom}} = \rho_\Lambda / \bar{M}_{\text{Pl}}^2$ ; the branch sets  $\rho_\Lambda = \lambda_\Lambda \bar{M}_{\text{Pl}}^4$ , so  $\Lambda_{\text{geom}} = \lambda_\Lambda \bar{M}_{\text{Pl}}^2$ , giving  $\bar{M}_{\text{Pl}}^2$  and then  $G_N = \lambda_\Lambda / (8\pi \Lambda_{\text{geom}}^{\text{obs}})$ ; restoring  $c^3/\hbar$  closes it.  $\square$

### Non-circularity ([A], essential)

The anchor must be the curvature  $\Lambda_{\text{geom}}^{\text{obs}}$  (units  $\text{m}^{-2}$ ), *not* the SI vacuum energy density  $\rho_\Lambda^{\text{obs}} = \Lambda_{\text{geom}} c^4 / (8\pi G_N)$ , which already contains  $G_N$ . The curvature follows from  $\Lambda_{\text{geom}}^{\text{obs}} = 3\Omega_\Lambda H_0^2 / c^2$ .

**Two independent anchors (crosscheck with teeth).** The metrology can be pinned in two ways that should agree:

1. *Electroweak anchor* (Paper 5): the dimensionless  $G_N v_{\text{phys}}^2 = 4.06717 \times 10^{-34}$  fixes  $G_N$  once  $v_{\text{phys}}$  is calibrated non-gravitationally.
2.  $\Lambda$ -curvature anchor (this note):  $G_N = \frac{3c^3}{32\pi^3\hbar\Lambda_{\text{geom}}^{\text{obs}}} e^{-2\alpha_\star^{-1}}$ .

With the example anchor  $H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $\Omega_\Lambda = 0.6847$  ( $\Lambda_{\text{geom}}^{\text{obs}} = 1.08914 \times 10^{-52} \text{ m}^{-2}$ ) the  $\Lambda$  anchor gives

$$G_N^{\text{TFPT}} = 6.65071 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \quad G_N^{\text{TFPT}} / G_N^{\text{CODATA}} - 1 = -3.5 \times 10^{-3}. \quad (39)$$

Closure algebra [I]; SI anchor and numerical match [P] (the cosmological anchor is model dependent; the Hubble tension applies). EW crosscheck: the  $\Lambda$ -pinned  $\bar{M}_{\text{Pl}} = 2.43964 \times 10^{18} \text{ GeV}$  gives  $v_{\text{geo}} = 242.6 \text{ GeV}$ , needing  $Z_{\text{EW}} = (246.22/242.61)^2 = 1.030$ .

### Prefactor branch ([A])

This note uses  $\Lambda_{\text{IR}} / \bar{M}_{\text{Pl}}^4 = \delta_{\text{top}} e^{-2\alpha_\star^{-1}}$ . A competing prefactor  $2c_3 e^{-2\alpha_\star^{-1}}$  would differ by  $2c_3 / \delta_{\text{top}} = 661.5$  and misscale  $G_N$  by the same factor; the branch must be versioned.

### Mandatory $\Lambda$ -layer typing ([A])

Three distinct  $\Lambda$  objects must never be conflated:

$$\underbrace{\frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}}_{\text{vacuum density quotient (dimensionless)}} \neq \underbrace{\frac{\Lambda_{\text{IR}}}{\bar{M}_{\text{Pl}}^4} = \delta_{\text{top}} e^{-2\alpha_\star^{-1}}}_{\text{seam-determinant / IR scale (dimensionless)}} \neq \underbrace{\Lambda_{\text{geom}}^{\text{obs}} = \frac{3\Omega_\Lambda H_0^2}{c^2}}_{\text{observed curvature } [\text{m}^{-2}]}.$$

Only  $\Lambda_{\text{geom}}^{\text{obs}}$  carries a dimension and is the non-circular anchor. Keeping the three typed protects the decuple bridge and the metrology closure from a spurious prefactor objection.

**Measurement-near (physical) decuple bridge.** Since  $v_{\text{phys}} = v_{\text{geo}} \sqrt{Z_{\text{EW}}}$  (Paper 5, charged-current residue, not a fit), the decuple bridge in the observable vev reads

$$\frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} \left[ \frac{v_{\text{phys}}}{5\beta_{\text{rad}}^2 \sqrt{Z_{\text{EW}}} \bar{M}_{\text{Pl}}} \right]^{10}. \quad (40)$$

This is the same identity, written in the low-energy charged-current scale rather than the geometric source scale; both forms should appear side by side. [I]/[P]

## 9 The landscape: $g_{\text{car}} = 5$ is overdetermined

Varying the carrier rank and rebuilding every constant from the same machinery ( $c_3$  fixed), the EM-closure value  $\alpha^{-1}(g_{\text{car}})$  is monotone and meets the observed value only at  $g_{\text{car}} = 5$ :

| $g_{\text{car}}$ | $N_{\text{fam}}$ | $\dim S^+$ | $\Omega_{\text{adm}}$ | $10b_1$   | $\alpha^{-1}(g_{\text{car}})$ | verdict                                             |
|------------------|------------------|------------|-----------------------|-----------|-------------------------------|-----------------------------------------------------|
| 3                | 2                | 4          | 8                     | 7         | 258.15                        | wrong $\alpha$ , wrong family count                 |
| 4                | 2.5              | 8          | 20                    | 17        | 187.31                        | non-integer families                                |
| <b>5</b>         | <b>3</b>         | <b>16</b>  | <b>48</b>             | <b>41</b> | <b>137.036</b>                | <b>SM, 3 families, observed <math>\alpha</math></b> |
| 6                | 3.5              | 32         | 112                   | 97        | 101.30                        | non-integer families                                |
| 7                | 4                | 64         | 256                   | 225       | 75.61                         | wrong $\alpha$                                      |

Several distinct requirements converge on  $g_{\text{car}} = 5$ : integer families ( $g_{\text{car}}$  odd); three families ( $(g_{\text{car}}+1)/2 = 3$ ; the observed  $\alpha^{-1}$  on the monotone curve; the SM structure  $(b, s) = (3, 2)$ ,  $\dim S^+ = 16$ ; and  $\Omega_{\text{adm}} = 48$ ,  $10b_1 = 41$ . A clean algebraic shadow:  $\frac{g(g+1)}{2} = 2^{g-1} - 1$  has  $g = 5$  as its non-trivial solution ([P]).

## 10 The $E_8$ carrier atlas — stated unambiguously

In the series  $E_8$  is a *downstream scale grammar*, not a primitive cause: the boundary kernel fixes  $\alpha$ , the carrier rank, families and admissibility without  $E_8$  upstream. Three levels.

### 10.1 Level 1 — original $E_8$ data of the series ([I])

$$|R(E_8)| = 248 - 8 = 240 = 5 \cdot 48 = 5 \Omega_{\text{adm}} = \text{lcm}(\Omega_{\text{adm}}, D_{\text{start}}) = \text{lcm}(48, 60), \quad (41)$$

$$D_{\text{start}} = g_{\text{car}} \cdot \dim \mathfrak{g}_{\text{SM}} = 5 \cdot 12 = 60, \quad \gcd(\Omega_{\text{adm}}, D_{\text{start}}) = 12 = \dim \mathfrak{g}_{\text{SM}}, \quad (42)$$

$$\frac{|R(E_8)|}{D_{\text{start}}} = \frac{240}{60} = 4 = \frac{\Omega_{\text{adm}}}{\dim \mathfrak{g}_{\text{SM}}}, \quad 2880 = D_{\text{start}} \Omega_{\text{adm}} = |R(E_8)| \cdot \dim \mathfrak{g}_{\text{SM}}, \quad (43)$$

$$\kappa_{E_8} = \frac{\gamma_{\text{car}}}{\ln(248/60)} = 0.58723, \quad X(n, r, I_1) = \frac{\bar{M}_{\text{Pl}}}{8} \sin^2 \theta_{13} \left( \frac{60 - 2n}{60} \right)^{\kappa_{E_8}} e^{-(8-r)/64} e^{-12\pi I_1}. \quad (44)$$

**Coxeter/gauge typing of the lcm–gcd pair.** Writing  $D_{\text{start}} = 60 = 2h_{E_8}$  and  $\Omega_{\text{adm}} = 48 = 6 \text{rank}(E_8)$ , the relations become  $\gcd(2h_{E_8}, 6r_{E_8}) = \dim \mathfrak{g}_{\text{SM}} = 12$  and  $\text{lcm}(2h_{E_8}, 6r_{E_8}) = |R(E_8)| = 240$ : the local SM algebra is the gcd of the Coxeter double-period and the  $\mathbb{Z}_6$  rank-occupancy, and the  $E_8$  root count is their lcm. Likewise the EW exponent reads  $I_1^{\text{EW}} = \frac{41}{48} = \frac{h_{E_8}}{36} + \frac{1}{6r_{E_8}} = \frac{5}{6} + \frac{1}{48}$  (Coxeter principal part plus a rank singlet). [I] (arithmetic) / [P] (typing).

### 10.2 Level 2 — new exact counting identities ([I])

$$|R(E_8)| = 240 = 16 \cdot 15 = \dim S^+ (\dim S^+ - 1), \quad (45)$$

$$\dim E_8 = 248 = 8 \cdot 31 = (g_{\text{car}} + N_{\text{fam}})(2^{g_{\text{car}}} - 1), \quad (46)$$

$$|R(E_8)| = 8(31 - 1) = \text{rank}(E_8)(2^{g_{\text{car}}} - 2), \quad \text{rank}(E_8) = 8 = g_{\text{car}} + N_{\text{fam}} = 5 + 3, \quad (47)$$

$$10b_1 = 41 = \frac{|R(E_8)|}{6} + 1, \quad I_1^{\text{EW}} = \frac{41}{48} = \gamma_{\text{car}} + \frac{1}{\Omega_{\text{adm}}}, \quad (48)$$

with  $31 = 2^{g_{\text{car}}} - 1$  the number of non-empty occupation words of the  $[5, 4, 2]$  carrier code.



### 10.3 Level 2.5 — the even-flip transition atlas (a falsifiable program)

Identify  $S^+$  with the even-weight binary code on five bits. A transition  $x \rightarrow y \neq x$  has flip  $F = x \oplus y$ , even and nonempty, so  $|F| \in \{2, 4\}$ ; per state there are  $\binom{5}{2} + \binom{5}{4} = 10 + 5 = 15$  flips, giving

$$240 = 16 \cdot (10 + 5) = \underbrace{160}_{|F|=2} + \underbrace{80}_{|F|=4}, \quad (49)$$

and the 120 undirected flips match the 120  $E_8$  root *pairs* ([I], counting). This makes the atlas conjecture *testable*: a genuine root map must send each flip to a norm-2 vector in  $\mathbb{R}^8$  with inner products in  $\{-2, -1, 0, 1, 2\}$  reproducing the  $E_8$  profile (per root: 1 self, 56 at +1, 126 orthogonal, 56 at -1, 1 anti).

#### Honest obstruction ([A])

The flip atlas splits  $160 + 80$  by flip size; the standard  $E_8$  ( $D_8$ ) split is  $112 + 128$ . Since  $160 + 80 \neq 112 + 128$ , no flip-size-preserving map is an  $E_8$  root system; a metrization, if it exists, must mix flip sizes. Until a norm-2,  $E_8$ -profile embedding is exhibited,  $E_8$  stays a *counting* atlas, not a derived root origin.

**Sharper two-stage audit (Coxeter before Weyl).** A full root-system isomorphism is hard; a much earlier, cheaper test is the Coxeter spectrum. *Stage A*: find a natural order-30 operator  $C_{\text{TFPT}}$  on the even-flip atlas (or its 8-dimensional primitive-character space) with characteristic polynomial  $\chi_{C_{\text{TFPT}}}(t) = \Phi_{30}(t)$ . *Stage B (only if A holds)*: the norm-2,  $E_8$ -profile, Weyl-closure metrization. Stage A can fail or carry far sooner than B, so it is the right first falsifier. [A]

#### Audit only — non-load-bearing speculation: the dim 496 heterotic count [A]

*This box feeds no derivation; it is an arithmetic coincidence kept for completeness only, under the no-free-pattern rule.* The even half  $S^+$  and odd half  $S^-$  of the exterior algebra each have dimension  $2^{g_{\text{car}}-1} = 16$  and each carry  $16 \cdot 15 = 240$  transitions. Together the full carrier ( $2^{g_{\text{car}}} = 32$  states) gives

$$\dim S^+(2^{g_{\text{car}}} - 1) = 16 \cdot 31 = 496 = \dim(E_8 \times E_8) = \dim SO(32) = \binom{32}{2} \quad (\text{count}),$$

i.e. the two halves read as two  $E_8$  root atlases, all  $\binom{32}{2} = 496$  pairs as  $SO(32)$  — the two anomaly-free heterotic gauge dimensions. The *dimension* equality is exact; that the carrier algebra *is* either gauge group is unproven and not used anywhere. [A] (audit-only).

### 10.4 Level 2.7 — the carrier Coxeter theorem (the strongest new bridge)

Use the universal root-system identity  $|R| = \text{rank} \cdot h$  (Coxeter number  $h$ ). With the carrier values  $\text{rank}(E_8) = g_{\text{car}} + N_{\text{fam}} = 8$  and  $|R(E_8)| = \dim S^+(\dim S^+ - 1) = 240$ ,

$$h_{E_8} = \frac{|R(E_8)|}{\text{rank}(E_8)} = \frac{240}{8} = 30 = N_{\text{fam}} \binom{g_{\text{car}}}{2} = N_{\text{fam}} A_{\Lambda} = 2g_{\text{car}} N_{\text{fam}} = 2^{g_{\text{car}}} - 2 \quad [\text{I}]. \quad (50)$$

The Coxeter number of  $E_8$  is exactly three families times the cosmological pair action: each family contributes one  $\Lambda$ -decuple ( $A_{\Lambda} = \binom{5}{2} = 10$ ) to the Coxeter period. Equivalently the action ladder is a Coxeter ladder,

$$A_H = \frac{h_{E_8}}{2N_{\text{fam}}} = 5, \quad A_{\Lambda} = \frac{h_{E_8}}{N_{\text{fam}}} = 10, \quad (51)$$

so Hubble reads half a Coxeter period per family and  $\Lambda$  a full Coxeter period per family. [P]

**Rank–Coxeter coupling.** Equivalently  $\dim S^+ = 2 \operatorname{rank}(E_8) = 16$  and  $h_{E_8} = 2(\dim S^+ - 1) = 2 \cdot 15 = 30$ , so  $\dim E_8 = (g_{\text{car}} + N_{\text{fam}})(h + 1) = 8 \cdot 31$ . The two counting forms  $16 \cdot 15 = 8 \cdot 30 = 240$  coincide because  $\dim S^+ = 2r$ . Dropping  $\nu^c$  sends  $\dim S^+ \rightarrow 15$ , which breaks *both*  $\dim S^+ = 2r$  and  $\dim S^+(\dim S^+ - 1) = 240$ :  $\nu^c$  is the state that closes the rank–Coxeter coupling, not an optional singlet. [\[I\]](#)/[\[P\]](#)

### 10.5 Level 2.8 — positive-root moment: $|R^+(E_8)| = \operatorname{Tr}_{S^+} X^2 = 5!$

The number of positive  $E_8$  roots equals the master hypercharge moment:

$$\boxed{|R^+(E_8)| = 120 = \operatorname{Tr}_{S^+} X^2 = 5!} \quad \text{[I]}, \quad (52)$$

so the second hypercharge moment of one family *is* the positive-root half-system. The whole integer chain then reads off  $|R^+|$ :

$$\sum_{f,j} L_{f,j} = \frac{|R^+(E_8)|}{3} = 40, \quad 10b_1 = 1 + \frac{|R^+(E_8)|}{3} = 41, \quad \delta_2 = 4! |R^+(E_8)| c_3^8. \quad (53)$$

The  $\mathbb{Z}_6$  slice is then  $|R(E_8)|/6 = \frac{|R^+|}{3} = 40 = \Omega_{\text{adm}}\gamma$ : one  $\mathbb{Z}_6$  phase slice of the  $E_8$  root atlas is exactly the full flavor transport budget. [\[I\]](#) (arithmetic) / [\[P\]](#) (interpretation).

### 10.6 Level 2.9 — the 2·3·5 Coxeter cyclotomic compiler (meta-theorem)

The three hard carrier atoms — weak rank 2, color rank / family count 3, carrier rank  $g_{\text{car}} = 5$  — generate the  $E_8$  data *cyclotomically*, without taking  $E_8$  as input. Set

$$\boxed{h := 2 \cdot 3 \cdot g_{\text{car}} = 30, \quad r := \varphi(h) = \varphi(30) = 8 = g_{\text{car}} + N_{\text{fam}},} \quad (54)$$

where  $h$  is the Coxeter number and  $r = \varphi(2 \cdot 3 \cdot 5) = 8$  the rank. Then the standard  $|R| = rh$ ,  $\dim = r(h+1)$  give

$$|R(E_8)| = rh = 8 \cdot 30 = 240, \quad \dim E_8 = r(h+1) = 8 \cdot 31 = 248, \quad h+1 = 2^{g_{\text{car}}} - 1 = 31. \quad (55)$$

Moreover the  $E_8$  exponents are exactly the primitive residues modulo  $h = 30$ :

$$\operatorname{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \quad \sum_{m \in (\mathbb{Z}/30\mathbb{Z})^\times} m = 120 = \operatorname{Tr}_{S^+} X^2 = |R^+(E_8)|, \quad (56)$$

so the second hypercharge moment equals the sum of the  $E_8$  exponents and the positive-root count. The  $\mathbb{Z}_6 \times \mathbb{Z}_5 \simeq \mathbb{Z}_{30}$  tact has  $\varphi(30) = 8$  primitive characters, whose phases  $e^{2\pi i m/30}$  are the Coxeter eigenphases of  $E_8$ ; its characteristic polynomial is the 30th cyclotomic polynomial

$$\chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1. \quad (57)$$

**Reading:**  $E_8$  here is not primarily a root count but the *Coxeter cyclotomic readout of the 2, 3, 5 carrier*. [\[I\]](#) (arithmetic) / [\[P\]](#) (interpretation).

### 10.7 Level 3 — conjectural geometric readings ([\[P\]](#))

**Conjecture 1** (Carrier transition atlas).  $R(E_8) \cong \{(a, b) \in S^+ \times S^+ : a \neq b\}$  as a structure-bearing transition set, since  $16 \cdot 15 = 240$ . *Open*: a structure-preserving map (roots, lengths, Weyl data), not only the count.

**Conjecture 2** (Least common refinement / rank-8 over the exhausted carrier).  $240 = \operatorname{lcm}(\Omega_{\text{adm}}, D_{\text{start}})$  reads  $E_8$  as the least common refinement of the occupancy lattice and the  $\mathbb{Z}_6$  phase lattice;  $248 = (g_{\text{car}} + N_{\text{fam}})(2^{g_{\text{car}}} - 1)$  reads it as a rank-8 structure over the 31 non-empty carrier words.



## Two routes to $E_8$ — keep their status separate

Levels 1–2 are exact arithmetic. There are now *two* distinct routes to  $E_8$ , and they must not be conflated:

- **Glue route (closed, [L]):**  $E_8 = (D_5 \oplus A_3) + \mu_4$ -glue is built explicitly in Section 3 (disc( $D_5$ ) = disc( $A_3$ ) =  $\mathbb{Z}_4$ , glue index 4, 240 roots of norm 2, simple-root Gram det 1). This closes the former “blind 240-metrisation”.
- **Direct flip-atlas route (open, [A], optional bonus):** the even-flip transition set  $240 = 160 + 80$  is *not* a flip-size-preserving  $E_8$  map, since  $160 + 80 \neq 112 + 128$ . A direct structure-preserving metrisation of the flip graph remains open — a nice-to-have, not load-bearing, because the glue route already delivers  $E_8$ .

So  $E_8$  is closed via the glue; only the direct flip atlas is open.

## 11 $\mathbb{Z}_6$ monodromy, flavor and CP readouts

A single  $\mathbb{Z}_6$  motif recurs in the quotient  $G_{\text{phys}} = (SU(3) \times SU(2) \times U(1)_Y) / \mathbb{Z}_6$ , the transport cusps  $\{1, (2/3)^6, (1/3)^6\}$ ,  $\delta_{\text{ph}} = z_\star^{1/6}$ , and the CP phases

$$\delta_{\text{CKM}}^{(0)} = \frac{\pi}{3}, \quad \delta_{\text{CKM}} = \frac{\pi}{3} + 3\lambda_C^2 = 1.19823 \text{ rad}, \quad \delta_{\text{CP}}^{\text{PMNS}} = \frac{4\pi}{3} = 240^\circ. \quad (58)$$

From the documented inputs  $s_{12} = \lambda_C$ ,  $s_{23} = u/(1 + \lambda_C)$ ,  $s_{13} = \lambda_C^3/3$  one gets the new derived readouts  $|V_{td}| = 0.00908$ ,  $|V_{ts}| = 0.04264$ ,  $J_{\text{CKM}} = 3.33 \times 10^{-5}$ . On the neutrino side, with  $\sin^2 \theta_{13} = e^{-5/6}u = 0.02311$ ,  $\sin^2 \theta_{23} = 0.4557$ ,  $\delta_{\text{CP}} = 4\pi/3$  and the solar candidate  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{u}{2} = 0.30675$ , a near-maximal  $|J_{\text{CP}}'| \approx 0.030$  (sharp readout for DUNE/Hyper-Kamiokande). [P]

### 11.1 CP as triple seed variance; a closed observable quadrilateral

The Cabibbo angle is the Bernoulli width of the seed. The *exact* CKM phase is the family holonomy of Paper 3,  $\delta_{\text{CKM}} = \arg \text{Tr} \Pi_\Delta(\nabla_F^\star)$ ; on the small-area branch this has the compact readout form (with an  $O(\lambda_C^4)$  remainder)

$$\lambda_C = \sqrt{u(1-u)}, \quad \boxed{\delta_{\text{CKM}} - \frac{\pi}{3} = N_{\text{fam}} u(1-u) = 3\lambda_C^2} \quad [\text{I}]/[\text{P}] \text{ (readout; exact = holonomy)}. \quad (59)$$

The coefficient 3 is the family number  $N_{\text{fam}}$ : weak CP is the  $\mathbb{Z}_6$  base angle  $\pi/3$  plus  $N_{\text{fam}}$  times the seed (Bernoulli) variance. The  $\mathbb{Z}_6$  base angle now has an  $A_3$  root: since  $\mathbb{Z}_6 = |R^+(A_3)| = 6$  (the positive-root hexagon of the family algebra), the CP phases are integer steps of the  $A_3$  holonomy quantum  $2\pi/|R^+(A_3)|$ ,

$$\boxed{\delta_{\text{CKM}} = \frac{2\pi}{|R^+(A_3)|} + N_{\text{fam}} u(1-u), \quad \delta_{\text{CP}}^{\text{PMNS}} = 4 \cdot \frac{2\pi}{|R^+(A_3)|} = \frac{4\pi}{3}} \quad [\text{I}]/[\text{P}]. \quad (60)$$

So weak CP = ( $A_3$  positive-root phase step) + (family count  $\times$  seed variance), while strong CP is the *determinant* sector, erased. The solar angle reads as a carrier root plus a half-seed,  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{u}{2} = -y_- - \frac{u}{2}$  (since  $-y_- = \frac{1}{3}$ ), so PMNS = carrier root + seed +  $A_3$  holonomy phase. [P] Strong CP is killed by the determinant selector ( $\arg \det M_u = \arg \det M_d = 0$ ,  $\theta = 0$ ), while weak CP survives as the seed-deformed  $\mathbb{Z}_6$  holonomy: CP is not forbidden, only *determinant* CP is. [P] — the strong-CP closure lives on the admissible QFT branch and is *conditional* on the analytic hypotheses of Paper 4 (reflection positivity, temperedness, clustering, mass gap, OS reconstruction); its status is therefore weaker than the discrete lattice identities, even though the

semantics “determinant CP forbidden, holonomy CP allowed” is robust. Since  $u = \Omega_b + \beta_{\text{rad}}$  exactly, eliminating the seed closes the four UV-shadow observables into one cycle:

$$\lambda_C^2 = (\Omega_b + \beta_{\text{rad}})(1 - \Omega_b - \beta_{\text{rad}}), \quad \sin^2 \theta_{13} = e^{-5/6}(\Omega_b + \beta_{\text{rad}}), \quad \Omega_b = \left(1 - \frac{1}{4\pi}\right)e^{5/6} \sin^2 \theta_{13}. \quad (61)$$

One measured member of  $\{\lambda_C, \theta_{13}, \Omega_b, \beta_{\text{rad}}\}$  fixes the other three. **[I]** (the four rows remain operationally distinct readout classes).

## 12 Mass hierarchy as a hexagon with one extra turn

The Paper-3 transport length packets are

$$L_u = (7, 3, 0), \quad L_d = (7, 5, 2), \quad L_e = (8, 5, 3), \quad \sum_{f,j} L_{f,j} = 40 = \Omega_{\text{adm}} \gamma_{\text{car}} = 10b_1 - 1. \quad (62)$$

Modulo 6 they lie on related triples of the  $C_6$  hexagon; the first generation carries one extra full turn ( $u, d : 1 + 6; e : 2 + 6$ ), costing  $\lambda_Y^6 = (u(1-u))^3 \approx 1.28 \times 10^{-4}$  and explaining why the first generation is so light — “one turn more”, not a free fit. **[P]** *Flagged proximity ([A], not exact)*: the full-turn cost is close to the topological seam defect,  $\lambda_C^6 / \delta_{\text{top}} = 1.061$  ( $\sim 6\%$ ), hinting that one flavor winding  $\approx$  one seam-defect unit  $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$  — a structural smell, not a theorem. Writing the length matrix  $L = R + W$  with  $R = L \bmod 6$  the residue shadow and  $W$  the full-winding part,

$$L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}}_R + 6 \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_W, \quad \sum R = 22, \quad \sum W = 18 = 3 \cdot 6, \quad \sum L = 40,$$

and the column sums are  $(22, 13, 5)$  with  $S_1 = 22 = \sum R$  and  $S_2 + S_3 = 18 = \sum W$ . So the first generation is light because it stores the *entire* residue budget plus the three full  $C_6$  windings; generations 2 and 3 are the winding complement. **[I]** (arithmetic).

### 12.1 Lepton flavor decuple: the charged-lepton determinant carries $u^{10}$

The Paper-3 source masses on the  $v_{\text{geo}}$  branch are  $(\hat{m}_e, \hat{m}_\mu, \hat{m}_\tau) = \frac{v_{\text{geo}}\pi}{\sqrt{2}} (\frac{16}{7}u^5, \frac{4}{3}u^3, \frac{7}{6}u^2)$ . Their product is a pure decuple in the seed:

$$\boxed{\frac{\hat{m}_e \hat{m}_\mu \hat{m}_\tau}{(v_{\text{geo}}\pi/\sqrt{2})^3} = \left(\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6}\right) u^{5+3+2} = \frac{32}{9} u^{10}} \quad \text{[I]}, \quad (63)$$

with  $5+3+2 = g_{\text{car}} + b + s = 10 = \binom{5}{2}$ . So the charged-lepton Yukawa powers  $(n_e, n_\mu, n_\tau) = (5, 3, 2)$  are the three carrier numbers  $(g_{\text{car}}, b, s)$ , and the lepton determinant carries the same decuple ( $u^{10}$ ) that the cosmological constant carries in the gauge sector ( $A_\Lambda = 10$ ):  $\Lambda$  and the charged-lepton determinant are two realisations of the same degree- $\binom{5}{2}$  sector.

**Sum–product reading.** The lepton powers  $(5, 3, 2)$  encode *both* decuple numbers at once: their sum is the  $\Lambda$  action, their product is the  $E_8$  Coxeter number,

$$5 + 3 + 2 = 10 = A_\Lambda, \quad 5 \cdot 3 \cdot 2 = 30 = h_{E_8} = 2 \cdot 3 \cdot 5, \quad (64)$$

and  $(5, 3, 2)$  is the prime factorisation of  $h_{E_8} = 30$ . So  $\det M_\ell$ ,  $\rho_\Lambda$  and the  $E_8$  Coxeter period read the same carrier core additively ( $= 10$ ) and multiplicatively ( $= 30$ ). **[I]/[P]**

## 12.2 Why the top Yukawa is $\approx 1$ : the $L = 0$ anchor (Froggatt–Nielsen ladder)

The transport length  $L_f$  is a Froggatt–Nielsen-type charge with expansion parameter  $\lambda_C$ :

$$\boxed{y_f \simeq \lambda_C^{L_f} \times \Lambda_f \ (\Lambda_f = \mathcal{O}(1)), \quad y_t \simeq 1 \text{ because } L_t = 0} \quad [\mathbf{I}]/[\mathbf{P}]. \quad (65)$$

The top sits at  $L_t = 0$  (the *unwound* carrier direction), so  $y_t = \Lambda_t \simeq 1$  ( $\sqrt{2} m_t/v = 0.991$ ); every other charged fermion is suppressed by integer powers of  $\lambda_C = 0.2244$ . Numerically (with  $\mathcal{O}(1)$  sector prefactors  $\Lambda_f \in [0.43, 1.07]$ ):

| $f$                   | $t$  | $b$  | $\tau$ | $c$  | $\mu$ | $s$  | $u, d$                 | $e$  |
|-----------------------|------|------|--------|------|-------|------|------------------------|------|
| $L_f$                 | 0    | 2    | 3      | 3    | 5     | 5    | 7                      | 8    |
| $y_f/\lambda_C^{L_f}$ | 0.99 | 0.48 | 0.90   | 0.65 | 1.07  | 0.94 | $\sim 0.4\text{--}0.9$ | 0.46 |

So the entire charged-fermion hierarchy is one  $\lambda_C$ -ladder anchored by the top at  $L = 0$ ; the lightness of the first generation ( $L = 7, 8$ ) and the heaviness of the top ( $L = 0$ ) are the two ends of the same integer winding spectrum. This is the standard Froggatt–Nielsen mechanism with  $\lambda_C$  as the flavon ratio and  $L_f$  as the FN charge — a clean bridge between TFPT transport and an established flavor mechanism.

## 13 Inflation interface (Paper 6)

The documented scalaron is Starobinsky  $R^2$ ; with the pivot  $N_\star = 57.1$ :

$$\begin{aligned} n_s &= 1 - \frac{2}{N_\star} = 0.96497, \quad r = \frac{12}{N_\star^2} = 0.00368, \\ \frac{M_{\text{scal}}}{\bar{M}_{\text{Pl}}} &= \sqrt{8\pi} c_3^4 = c_3^{7/2} = (8\pi)^{-7/2} \Rightarrow M_{\text{scal}} \approx 3.06 \times 10^{13} \text{ GeV}. \end{aligned} \quad (66)$$

$n_s$  matches Planck;  $r = 0.0037$  is the falsifiable target for LiteBIRD/CMB-S4.

**Amplitude (status-clean).** The scalaron mass is a pure seam power,

$$\boxed{\frac{M_{\text{scal}}^2}{\bar{M}_{\text{Pl}}^2} = c_3^{r-1} = c_3^{\Omega_{\text{adm}} - 10b_1} = c_3^7} \quad [\mathbf{I}], \quad (67)$$

an *exact* identity (the exponent  $7 = 48 - 41$  is the post-flavor, post-Higgs occupancy deficit). Paper 6 reduces the geometric gravity action to an FRW  $R^2$ /scalaron branch in which  $M_{\text{scal}}$  is the Starobinsky mass  $M$ ; the identification is therefore derived in that sector, not posited. With the standard large- $N$  amplitude  $A_s = \frac{N_\star^2}{24\pi^2} (M/\bar{M}_{\text{Pl}})^2$  this gives

$$A_s = \frac{N_\star^2 c_3^7}{24\pi^2} = \frac{N_\star^2 \delta_2}{8640\pi} = 2.17 \times 10^{-9} \quad [\mathbf{P}]/[\mathbf{A}], \quad (68)$$

matching Planck ( $\simeq 2.1 \times 10^{-9}$ );  $\delta_2 = 2880 c_3^8$  is the second seam defect. The seam-power form is exact  $[\mathbf{I}]$ ; the amplitude is then determined *within* the  $R^2$  sector, the only residual being the geometric-gravity (Hodge) closure on which that sector rests  $[\mathbf{P}]$ . *Convention note:*  $A_s \propto N_\star^2$ , so the quoted  $2.17 \times 10^{-9}$  uses the pivot  $N_\star = 57.1$ ; the alternative seed pivot  $N_\star = 3/u - 1 = 55.4$  gives  $2.05 \times 10^{-9}$ . A number should only be quoted with  $N_\star$  fixed.

## 14 Further simple bridges

- **Dark-energy equation of state  $w = -1$  (exact).** The TFPT  $\Lambda$  is a genuine vacuum constant  $\rho_\Lambda = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}} \bar{M}_{\text{Pl}}^4$ , not a rolling field, so  $w = -1$  identically (no quintessence). A robust measurement  $w \neq -1$  would falsify the  $\Lambda$ -branch.  $[\mathbf{I}]$  (structural).

- **Effective neutrino number**  $N_{\text{eff}} \simeq 3$ . Three families ( $N_{\text{fam}} = 3$ ) give three light neutrinos and the standard  $N_{\text{eff}} \simeq 3.04$ . [I].
- **Top Yukawa**  $\simeq 1$  as the  $L = 0$  anchor (above): a bridge to the Froggatt–Nielsen flavor mechanism.
- *Flagged (not clean)*:  $m_H/v = 0.509$  is near  $\frac{1}{2}$  (1.7%);  $m_W/m_Z = 0.881$  matches the running  $\sqrt{1 - \sin^2 \theta_W(M_Z)}$  (not the tree  $\sqrt{5/8} = 0.79$ ). No clean closed bridge for the Higgs/ $W/Z$  masses yet. [A]

## 15 Closures that the architecture rests on

Five results close or sharpen the discrete core. They are *proved in this document set* — the  $\varphi_0^{\text{ret}}$ -ladder, the H2 realisation and the strong-CP/dynamics gap in document 2 (tfpt\_2\_standard\_model), the  $E_6 \times A_2$  shadow in document 3 (tfpt\_3\_e8\_audit\_bootstrap) — and are collected here because the architecture above depends on them.

1. **The  $\varphi_0^{\text{ret}}$ -ladder (whole spectrum, one formula).** Every fermion mass is  $\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}$  with  $\lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}$ , and the closed evaluation gives the universal form  $\lambda_Y^L \Lambda = \pi c (\varphi_0^{\text{ret}})^k$  — the nine  $\Lambda$  are hexagonal-resolvent outputs (not free numbers); the *charged-lepton* ones are exact, the *quark* ones are reduced to the ( $U_{\text{wall}}$ ) selector. The  $\varphi_0^{\text{ret}}$ -order ladder is  $t(0) > \{b, c, \tau\}(2) > \{s, \mu\}(3) > \{u, d\}(4) > e(5)$ . [I] (leptons exact, quark ratios closed v49/v71) / [A] (absolute quark scale)
2. **H2 reduced via the Mehta–Seshadri framework.** Mehta–Seshadri supplies the equivalence (stable parabolic  $\leftrightarrow$  irreducible unitary); the TFPT-specific step — “stable = transport-minimal” and the geodesic  $C_6$  word length — is conditional on the unitarity (U) of  $\nabla_F^*$  (Paper 3) and the parabolic-degree  $\rightarrow C_6$  dictionary. H2 is the sharp realisation statement: the  $D_4$ -equivariant stable structure on  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  realises the branch with  $\det R = h(D_5) = 8$ , principal 2-minors (2, 3, 5). [I] (given (U))
3. **Strong CP / dynamics via the explicit gap.**  $\theta_{\text{eff}} = 0$  needs only  $\gamma_5$ -Hermiticity + polar mass + sheet involution + reflection positivity (no mass gap). The full dynamics closes on the physical sector because the  $C_6$  transport transfer operator has the *explicit* spectrum  $\{(k/3)^6\}$ , giving the mass gap  $\Delta = 6 \log \frac{3}{2} = 2.43$  in closed form  $\Rightarrow$  OS reconstruction  $\Rightarrow$  measure on the admissible sector. [I] (phys. sector)
4.  **$E_6 \times A_2$  flavor shadow.** The residue matrix sits inside an  $E_8$  secondary branching:  $248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$ , with  $\|R\|_F^2 = 78 = \dim E_6$ ,  $\det R = 8 = \dim A_2$ . The seven  $E_8$  slices form an audit raster (atlas note). [A]
5. **SM completeness.** The discrete structure and dimensionless core are closed; the dimensionful EW/QCD masses ( $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ ) are schema-level via  $\mathcal{R}_{\text{SM}}$ , with  $m_H$  resting on the closed UV quartic  $\lambda_\Phi = 1/(16\pi^2)$  (near-criticality). The only genuinely open SM point is the *exact* solar angle  $\theta_{12}$  (leading  $1/N_{\text{fam}} = \frac{1}{3}$ ). [P]

## 16 Status summary and open audit points

| Statement                                                                             | Status | Note                         |
|---------------------------------------------------------------------------------------|--------|------------------------------|
| Dual-root dictionary ( $Y$ , cusps, $\gamma_{\text{car}}$ , $\varphi_{\text{base}}$ ) | [I]    | exact, from $\{1/2, -1/3\}$  |
| $10b_1 = 5^2 + 4^2 = \Omega_{\text{adm}}\gamma + 1$                                   | [I]    | Pythagorean carrier identity |
| $I_1^{\text{EW}} = 41/48 = \text{Var}(Y) b_1$                                         | [I]    | exact                        |
| $\delta_2 = 4! \cdot 5! c_3^8$ , $\text{Tr } X^2 = 5!$                                | [I]    | factorial traces             |
| $5/4 = g_{\text{car}}/(g_{\text{car}} - 1)$ universal compression                     | [I]    | exact                        |
| $u = \frac{4}{3}c_3 + 48c_3^4$ ; one-decoder identities                               | [I]    | exact                        |

|                                                                                                                                                                               |             |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------|
| $\xi = \frac{3}{4}/(1 + 9/128\pi^3)$                                                                                                                                          | [I]         | corrected tree value                              |
| Ladder $A_{EW} : \mathcal{A}_H : \mathcal{A}_\Lambda = 1 : g_{\text{car}} : 2g_{\text{car}}$                                                                                  | [I]/[P]     | Hubble rung is a corollary                        |
| Ladder $= \binom{g_{\text{car}}}{0} : \binom{g_{\text{car}}}{1} : \binom{g_{\text{car}}}{2} ; 2g_{\text{car}} = \binom{g_{\text{car}}}{2} \Leftrightarrow g_{\text{car}} = 5$ | [I]         | new overdetermination of $g_{\text{car}} = 5$     |
| 2D action lattice $S_x = a(\alpha_\star^{-1}/g_{\text{car}}) + bL_0$                                                                                                          | [I]/[P]     | two integer labels per scale                      |
| <b>Decuple bridge</b> $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} [v/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})]^{10}$                    | [I]         | exact, $10 = 2g_{\text{car}} = \dim \Lambda^2 E$  |
| Family = ladder = Pascal row $\{1, 5, 10\}$ ; $16 = 1+5+10$                                                                                                                   | [I]         | central new identification                        |
| $g_{\text{car}} = 5$ via $2^{g_{\text{car}}-1} = \binom{g_{\text{car}}}{0} + \binom{g_{\text{car}}}{1} + \binom{g_{\text{car}}}{2}$                                           | [I]         | Pascal closure (unique)                           |
| $\text{Tr}_{S^+} X^2 = 5!$ generates 40, 41, 240, $\delta_2$                                                                                                                  | [I]         | master moment                                     |
| $ R(E_8)  = 16 \cdot 5 \cdot 3$ ; $\dim E_8 = 240 + 8$                                                                                                                        | [I]         | $E_8$ compression                                 |
| Even-flip atlas $240 = 160 + 80$                                                                                                                                              | [I]/[A]     | $E_8$ metrization open ( $\neq 112 + 128$ )       |
| $\delta_{\text{CKM}} - \pi/3 = 3u(1-u)$                                                                                                                                       | [I]         | CP = triple seed variance                         |
| Observable quadrilateral ( $u = \Omega_b + \beta_{\text{rad}}$ )                                                                                                              | [I]         | closed UV-shadow cycle                            |
| Power tower $v=x^1, H=x^5 R_H, \Lambda=x^{10} R_\Lambda$                                                                                                                      | [I]         | $K_5$ vertices/edges                              |
| $496 = 16 \cdot 31 = \dim(E_8 \times E_8) = \dim SO(32)$                                                                                                                      | [I]/[P]     | heterotic-dimension count                         |
| $A_{EW} = \alpha_\star^{-1}/\sqrt{\Delta_Y}, \Delta_Y = (b+s)^2 = 25$                                                                                                         | [I]         | discriminant action theorem                       |
| $\hat{m}_e \hat{m}_\mu \hat{m}_\tau \propto \frac{32}{9} u^{10}$ ; $(5, 3, 2) = (g_{\text{car}}, b, s)$                                                                       | [I]         | lepton flavor decuple                             |
| $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2}$                                                                                                                             | [I]         | scalaron seam power                               |
| $\delta_{\text{CKM}} - \pi/3 = N_{\text{fam}} u(1-u)$                                                                                                                         | [I]         | weak CP = $N_{\text{fam}} \times$ seed variance   |
| $L = R + W$ : $\sum R = 22, \sum W = 18$                                                                                                                                      | [I]         | first generation = residue store                  |
| $ R(E_8) /6 = \sum L = 40$ ( $\mathbb{Z}_6$ slice)                                                                                                                            | [I]         | one $\mathbb{Z}_6$ slice = flavor budget          |
| $h_{E_8} =  R /\text{rank} = 30 = N_{\text{fam}} \binom{5}{2} = N_{\text{fam}} A_\Lambda$                                                                                     | [I]/[P]     | carrier Coxeter theorem                           |
| $ R^+(E_8)  = \text{Tr}_{S^+} X^2 = 120 = 5!$                                                                                                                                 | [I]         | positive-root moment                              |
| common decuple $\binom{5}{2} = A_\Lambda = u^{5+3+2} = h_{E_8}/N_{\text{fam}} = 10$                                                                                           | [I]/[P]     | one carrier pair-decuple                          |
| $\lambda_C^6/\delta_{\text{top}} = 1.061$ (winding $\approx$ seam defect)                                                                                                     | [A]         | flagged proximity (not exact)                     |
| $y_f \simeq \lambda_C^{L_f}$ , top $L_t = 0 \Rightarrow y_t \simeq 1$                                                                                                         | [I]/[P]     | Froggatt–Nielsen ladder                           |
| $w = -1$ (true $\Lambda$ ); $N_{\text{eff}} \simeq 3$                                                                                                                         | [I]         | dark energy / neutrino species                    |
| 3-generator lattice $\{\alpha_\star^{-1}, L_0, L_c\}$ ; $M_{\text{scal}} = e^{-\frac{7}{2}L_c}$                                                                               | [I]         | seam axis $L_c = \ln(1/c_3)$                      |
| $\dim S^+ = 2 \text{rank}(E_8), h_{E_8} = 2(\dim S^+ - 1) = 2^9 - 2$                                                                                                          | [I]         | rank–Coxeter coupling ( $\nu^c$ closes it)        |
| $\gcd(2h, 6r) = \dim \mathfrak{g}_{\text{SM}} = 12, \text{lcm} = 240$                                                                                                         | [I]/[P]     | SM algebra = gcd; $E_8$ count = lcm               |
| $I_1^{\text{EW}} = h_{E_8}/36 + 1/(6r) = 41/48$                                                                                                                               | [I]/[P]     | Coxeter part + rank singlet                       |
| $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^{-1} = c_3^7$ exact; $A_s = N_\star^2 c_3^7/24\pi^2$                                                                           | [I]/[P]/[A] | seam power exact; $A_s$ conditional (Starobinsky) |
| $h_{E_8} = 2 \cdot 3 \cdot 5 = 30, r = \varphi(30) = 8$                                                                                                                       | [I]/[P]     | 2, 3, 5 Coxeter cyclotomic compiler               |
| $\text{Exp}(E_8) = (\mathbb{Z}/30)^\times, \sum = 120, \chi_C = \Phi_{30}$                                                                                                    | [I]/[P]     | exponents = primitive residues                    |
| lepton $(5, 3, 2)$ : $\sum = 10 = A_\Lambda, \prod = 30 = h_{E_8}$                                                                                                            | [I]/[P]     | sum=action, product=Coxeter                       |
| Hubble $H_0/\bar{M}_{\text{Pl}} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$                                                                                         | [I]         | exact corollary                                   |
| $\Lambda$ -anchored closure $G_N = \frac{3c^3}{32\pi^3 \hbar \Lambda_{\text{geom}}} e^{-2\alpha_\star^{-1}}$                                                                  | [I]/[P]     | $G_N$ to $-0.35\%$                                |
| $g_{\text{car}} = 5$ overdetermined                                                                                                                                           | [P]         | monotone $\alpha^{-1}(g_{\text{car}})$            |
| $ R(E_8)  = 240 = 16 \cdot 15, 248 = 8 \cdot 31$                                                                                                                              | [I]         | counting identities                               |
| $E_8 \leftrightarrow$ carrier-transition map                                                                                                                                  | [P]         | needs structure map                               |
| $\delta_{\text{CP}} = 4\pi/3, \theta_{12} = \frac{1}{3} - \frac{u}{2}$                                                                                                        | [P]         | PMNS readouts                                     |
| $ V_{td} ,  V_{ts} , J_{\text{CKM}}, J_{\text{CP}}^\nu; n_s, r$                                                                                                               | [P]         | derived readouts                                  |

**Open audit points:** (1) *discharged* — the EM closure  $F_{U(1)}(\alpha) = 0$  is now stated explicitly (Grammar III), with existence/uniqueness and root  $\alpha_\star^{-1} = 137.0359992168$  ( $2.9 \times 10^{-10}$ ,  $1.9\sigma$  vs CODATA-2022); (2) keep  $\gamma_{\text{car}} = 5/6$  vs  $\gamma_E$  distinct; (3) type  $M_{\text{Pl}}$  vs  $\bar{M}_{\text{Pl}}$  and “pure” ( $\mathcal{A}_\Lambda = 274.07$ ) vs “total” ( $-\ln(\delta_{\text{top}} e^{-2\alpha_\star^{-1}}) = 283.10, -\ln(\rho_\Lambda/\bar{M}_{\text{Pl}}^4) = 276.65$ ) actions consistently; (4) version the  $\Lambda$  prefactor branch; (5) the scalaron-mass  $\rightarrow$  Starobinsky- $M$  identification is the *only* condition

for  $A_s = N_\star^2 c_3^7 / (24\pi^2)$  (the seam power  $M_{\text{scal}}^2 / \bar{M}_{\text{Pl}}^2 = c_3^7$  itself is exact); (6) the  $E_8$  lattice is now built explicitly as  $(D_5 \oplus A_3) + \mu_4\text{-glue}$  (Section 3), so the former “blind 240-metrisation” is closed. The flavor status (companion parabolic-weight note) now splits cleanly: the *anchor multiset*  $\{1, 1, 2\}$  is derived (Schur–Horn + stability) [I]; the *sector assignment*  $(1, 1, 2)$  follows from the  $D_5$  cusp ordering [I]/[P]; the *first-generation* +6 *winding* is the hierarchy convention [P]; the six *non-anchor residues* are then forced (residue sets =  $D_6$  orbit of  $\{0, 1, 3\}$ , minus anchor, monotone) [I]. The single remaining geometric input is (H2), the specific  $D_6$  cusp labelling on  $\mathbb{P}^1 \setminus \mu_4$  — sharply characterised by the residue-matrix spectral selector ( $\det R = h(D_5)$ , principal minors  $(2, 3, 5)$ ), with a naive cusp-position shortcut explicitly ruled out [P]. The explicit Riemann–Hilbert construction confirms the rigid  $SU(3)$  system and its  $A_3$  Coxeter rotation; the genuine open step is the global parabolic-weight (Hodge) realisation of that selector.

**Honest negatives (reported to keep the discipline visible).** The TFPT lepton ladder gives a Koide ratio  $Q = (\sum m_\ell) / (\sum \sqrt{m_\ell})^2 = 0.6645$ , close to but *not* the empirical  $2/3 = 0.66667$  ( $-0.33\%$ ); TFPT is Koide-*like* but does not reproduce the precise relation. The baryon asymmetry  $\eta_B \approx 6 \times 10^{-10}$  and the proton/electron ratio  $m_p/m_e = 1836$  do *not* fall on a clean action charge or  $L_0$  ladder; they remain outside the present grammar. These are stated so the strong items are not read as a uniform success.

## 17 Consolidated picture

### The three nested grammars

**Carrier:**  $3 + 2 \Rightarrow \{y_-, y_+\} = \{-\frac{1}{3}, \frac{1}{2}\} \Rightarrow Y, \dim S^+ = 16, N_{\text{fam}} = 3, \Omega_{\text{adm}} = 48, b_1 = \frac{41}{10},$

**Seed:**  $u = \frac{4}{3}c_3 + 48c_3^4 \Rightarrow \{\lambda_C, \beta_{\text{rad}}, \Omega_b, \sin^2 \theta_{13}\},$

**Action:**  $\alpha_\star^{-1} \Rightarrow A_{\text{EW}} = \frac{\alpha_\star^{-1}}{g_{\text{car}}}, \mathcal{A}_\Lambda = 2\alpha_\star^{-1}, \mathcal{A}_H = \alpha_\star^{-1} \Rightarrow \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} \left[ \frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}} \right]^{10}.$

A small grammar generates many sectors: the Standard-Model packet, flavor, the electroweak hierarchy, the cosmological constant, the metrology, and the inflation/CMB interface. The strongest new statements are exact: the decuple bridge ties  $\Lambda$  to the tenth power of the stripped electroweak scale with  $10 = 2g_{\text{car}}$ ; the Hubble scale is the square root of  $\Lambda$ ; and the Einstein normaliser is the corrected tree value  $\frac{3}{4}/(1 + 9/128\pi^3)$ . None of this is claimed as a proof of a final theory; it is offered as a tightening of the existing architecture for your assessment.

### The compiler in one line

TFPT = even carrier code + seed decoder + 2·3·5 Coxeter compiler

From the three atoms  $(2, 3, g_{\text{car}}=5)$ :

$$h_{E_8} = 2 \cdot 3 \cdot 5 = 30, \quad r_{E_8} = \varphi(30) = 8 = g_{\text{car}} + N_{\text{fam}},$$

$$|R(E_8)| = rh = 240, \quad \dim E_8 = r(h+1) = 8 \cdot 31 = 248,$$

$$\text{Exp}(E_8) = (\mathbb{Z}/30)^\times, \quad \sum \text{Exp} = 120 = \text{Tr}_{S^+} X^2 = |R^+|, \quad \chi_C = \Phi_{30},$$

$$\dim S^+ = 1+10+5 = 16 = 2r, \quad A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = 1:5:10, \quad \mathcal{A}_H = \frac{h}{6}, \mathcal{A}_\Lambda = \frac{h}{3}, N_{\text{fam}} = \frac{h}{A_\Lambda} = 3,$$

$$\sum L = \frac{|R^+|}{3} = 40, \quad 10b_1 = 1 + \frac{|R^+|}{3} = 41 = 1 + h \frac{\dim S^+}{\dim \text{gSM}},$$

$$\det M_\ell \propto u^{10}, \quad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} \propto \left[ \frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}} \right]^{10}, \quad A_{\text{EW}} = \frac{\alpha_\star^{-1}}{\sqrt{\Delta_Y}}, \quad \sqrt{\Delta_Y} = g_{\text{car}} = 5.$$

**The common decuple (= 10):**  $\binom{g_{\text{car}}}{2}$  (carrier pairs) =  $A_\Lambda$  (cosmological action) =  $u^{5+3+2}$  (lepton determinant) =  $h_{E_8}/N_{\text{fam}}$  (Coxeter per family).  $\Lambda$ , the charged-lepton determinant, the  $E_8$  Coxeter structure and the action ladder all read the *same* carrier pair-decuple. The open core is one question: can the even-flip atlas be metrised as a genuine  $E_8$  root system (the 160+80 vs 112+128 obstruction)?



## Part II

# The Coxeter–cyclotomic compiler and the explicit $E_8$ construction

### Abstract

This short companion compresses the TFPT 4.5 bridge layer to a single discrete object. The claim is that the Standard-Model family, the cosmological action ladder, the  $E_8$  counting data, the flavor budget, inflation and the cosmological constant are *not* independent coincidences but projections of one structure: the even-weight code  $C^+$  on  $g_{\text{car}} = 5$  slots, together with its  $\mathbb{Z}_{30}$  Coxeter spectrum and the single seam seed  $c_3 = 1/(8\pi)$ . The decisive new result is that  $C^+$  is precisely the half-spin weight orbit of  $D_5 = \mathfrak{so}_{10}$  ( $\text{Aut}(C^+) = W(D_5)$ , order 1920), that the family geometry  $\mathbb{P}^1 \setminus \mu_4$  is an  $A_3 = \mathfrak{su}_4$ , and that  $E_8$  is then the standard *closure algebra* of  $D_5 \times A_3$  — not an input. The fusion is exact:  $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ , so the four family corners  $\mu_4$  are precisely the glue code that closes  $D_5 \oplus A_3$  into the unimodular  $E_8$  lattice. The glue is metric: the discriminant-form norms  $q(D_5) = \frac{5}{4}$  and  $q(A_3) = \frac{3}{4}$  sum to the  $E_8$  root norm 2 — and these are exactly the pre-existing TFPT constants  $\delta_2/\delta_{\text{top}}^2 = \frac{5}{4}$  and  $\xi_{\text{tree}} = \frac{3}{4}$ . The  $A_3$  side is explicit: the residue pairing on  $H^1(\mathbb{P}^1 \setminus \mu_4)$  is the  $A_3$  Cartan form, and the Paper-2 corner rotation  $z \mapsto iz$  acts as the  $A_3$  Coxeter element (order 4, spectrum  $\{i, -1, -i\}$ ). The lift is no longer a count: the 240  $E_8$  roots are *explicitly* built from  $D_5 \oplus A_3$  plus the spinor ( $\mu_4$ ) glue, all of norm 2, Cartan determinant 1. Seven statements carry it; every lattice identity below is checked exactly, and the analytic metric is computed and characterised (a  $D_4$ -equivariant deformation), so no *lattice* structural step is left open in the  $D_5 \times A_3$  glue route (the upstream boundary kernel, EM closure and flavor Hecke saturation remain separate open problems). Master statement: *five carrier slots give  $D_5$ , four family corners give  $A_3$ , and  $\mu_4$  glues  $D_5 \times A_3$  into the  $E_8$  atlas.*

### The compression

$$\begin{aligned} \text{TFPT} &= \underbrace{D_5 \text{ half-spin carrier}}_{C^+} + \underbrace{A_3 \text{ four-point family}}_{\mathbb{P}^1 \setminus \mu_4} + c_3 \text{ seam seed}, \\ E_8 &= \text{closure of } D_5 \times A_3 \end{aligned}$$

with  $c_3 = \frac{1}{8\pi}$ ,  $g_{\text{car}} = 5$ ,  $h := 2 \cdot 3 \cdot g_{\text{car}} = 30$ ,  $r := \varphi(h) = \varphi(30) = 8$ ,  $N_{\text{fam}} = 3$ ,  $N_{\Phi} = 1$ ,  $\dim \mathfrak{g}_{\text{SM}} = 12$ . Equivalently  $C^+ + \mathbb{Z}_{30}$  Coxeter compiler +  $c_3$  seed; the  $D_5 \times A_3$  reading is the structural lift of the same object.

## Theorem 1 — the even carrier code

Let  $E$  be the rank- $g_{\text{car}} = 5$  carrier, split  $E = E_3 \oplus E_2$  (color  $b = 3$ , weak  $s = 2$ ). The one-generation spinor packet is the even exterior algebra, equivalently the even-weight binary code

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E \cong C^+ = \{x \in \mathbb{F}_2^5 : |x| \text{ even}\}, \quad \dim S^+ = |C^+| = 2^{g_{\text{car}}-1} = 16. \quad (69)$$

The two carrier roots  $y_+ = \frac{1}{2}$ ,  $y_- = -\frac{1}{3}$  generate all hypercharges ( $Y(Q_L) = y_+ + y_- = \frac{1}{6}$ ,  $Y(u^c) = 2y_-$ ,  $Y(e^c) = 2y_+$ ,  $Y(\nu^c) = 0$ ) and the transport cusps  $\{1, \frac{2}{3}, \frac{1}{3}\} = \{2y_+, -2y_-, 2(y_+ + y_-)\}$ , with  $\gamma_{\text{car}} = \text{Tr}_E Y^2 = 3y_-^2 + 2y_+^2 = \frac{5}{6}$ . [I]

## Theorem 2 — Exterior–Action Duality

On  $C^+$  the flip  $F = x \oplus y$  is even, so  $|F| \in \{0, 2, 4\}$ . The three relations  $R_0, R_2, R_4$  form a symmetric association scheme (the even Hamming scheme on five bits) with

$$\boxed{\begin{array}{l} \text{valencies } (1, 10, 5) = (\dim \Lambda^0 E, \dim \Lambda^2 E, \dim \Lambda^4 E), \\ \text{spectral multiplicities } (1, 5, 10) = A_{EW} : \mathcal{A}_H : \mathcal{A}_\Lambda \end{array}} \quad [\mathbf{I}]. \quad (70)$$

The eigenvalues (computed directly) are  $A_2 : \{10^{(1)}, 2^{(5)}, (-2)^{(10)}\}$  and  $A_4 : \{5^{(1)}, (-3)^{(5)}, 1^{(10)}\}$ , with  $A_2 + A_4 = K_{16} - \mathbf{1}$ . The two graphs are named objects:

$$\begin{array}{l} R_2(C^+) = \text{folded/complement, SRG}(16, 10, 6, 6) \text{ (the Clebsch complement);} \\ \boxed{R_4(C^+) = \text{Clebsch graph, SRG}(16, 5, 0, 2)} \quad [\mathbf{I}]. \end{array} \quad (71)$$

**Remark.** The Standard-Model family is the *valency* side  $(1, 10, 5)$  of one scheme; the cosmological action ladder is its *spectral* side  $(1, 5, 10)$ . The action ladder is therefore not merely “also binomial” — it is the spectral dual of the even exterior family. [P]

**The full Bose–Mesner algebra** (machine-checkable) fixes the duality beyond eigenvalues:

$$A_2^2 = 10 \mathbf{1} + 6A_2 + 6A_4, \quad A_2 A_4 = 3A_2 + 4A_4, \quad A_4^2 = 5 \mathbf{1} + 2A_2, \quad (72)$$

with eigenmatrix and multiplicity orthogonality

$$P = \begin{pmatrix} 1 & 10 & 5 \\ 1 & 2 & -3 \\ 1 & -2 & 1 \end{pmatrix}, \quad \det P = -64, \quad \boxed{P^\top \text{diag}(1, 5, 10) P = \text{diag}(16, 160, 80)} \quad [\mathbf{I}], \quad (73)$$

so the flip split  $240 = 160 + 80$  ( $|F| = 2, 4$ ) appears directly in the scheme’s eigenmatrix orthogonality.

**Carrier** =  $D_5$  (**verified**). The 16 vertices of  $C^+$  are exactly the weights of the  $D_5 = \mathfrak{so}_{10}$  half-spinor  $\frac{1}{2}(\pm 1, \dots, \pm 1)$  with an even number of minus signs. The flip-2 and flip-4 graphs are translation- and permutation-invariant, so the *affine* group  $C^+ \rtimes S_5$  (translations  $\rtimes$  coordinate permutations) acts by graph/scheme automorphisms; the group closure gives the automorphism group of the even Hamming scheme (equivalently the Clebsch graph),

$$\boxed{\text{Aut}_{\text{aff}}(C^+) = C^+ \rtimes S_5, \quad |\text{Aut}_{\text{aff}}(C^+)| = 2^{g_{\text{car}}-1} g_{\text{car}}! = 16 \cdot 120 = 1920 = |W(D_5)|} \quad [\mathbf{I}], \quad (74)$$

and  $1920 = r |R(E_8)| = 8 \cdot 240$ . (The *linear* code automorphism group alone is only the coordinate group  $S_5$  of order 120; the factor  $2^{g_{\text{car}}-1} = 16$  is the translation/scheme part. It is the affine/scheme group that equals  $W(D_5)$ .) Edges of the Clebsch graph:  $|E(R_4)| = \frac{16 \cdot 5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$  (the entire flavor transport budget). So the five-slot even code is the half-spin weight structure of  $\mathfrak{so}_{10}$ , the classical home of one SM family with  $\nu^c$ .

## Theorem 3 — the Coxeter closure equation selects $g_{\text{car}} = 5$

Requiring the even family to equal the action truncation through pair degree,  $\dim S^+ = 2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ , i.e.

$$\boxed{2^g - 2 = g(g+1) \iff g = 5} \quad [\mathbf{I}], \quad (75)$$

the unique positive-integer solution. At  $g = 5$  the two sides equal 30, so *carrier exhaustion*  $2^g - 2$  and the *Coxeter period*  $g(g+1)$  coincide only at five slots:  $2^{g_{\text{car}}} - 2 = 30 = h$  and  $g_{\text{car}}(g_{\text{car}} + 1) = 30$ .

## Theorem 4 — the $\mathbb{Z}_{30}$ cyclotomic Coxeter compiler

From the three atoms  $(2, 3, g_{\text{car}})$ , with  $h = 2 \cdot 3 \cdot g_{\text{car}} = 30$  and  $r = \varphi(h) = 8$ :

$$\boxed{|R(E_8)| = rh = 240, \quad \dim E_8 = r(h+1) = 8 \cdot 31 = 248, \quad h+1 = 2^{g_{\text{car}}} - 1 = 31} \quad \text{[I]}. \quad (76)$$

The  $E_8$  exponents are the primitive residues modulo 30,

$$\text{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \quad \chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1, \quad (77)$$

and  $|(\mathbb{Z}/30\mathbb{Z})^\times| = \varphi(30) = 8 = r$ . The local SM algebra and the root atlas are the two arithmetic projections of  $(r, h) = (8, 30)$ :

$$\Omega_{\text{adm}} = 6r = 48, \quad D_{\text{start}} = 2h = 60, \quad \gcd(6r, 2h) = \dim \mathfrak{g}_{\text{SM}} = 12, \quad \text{lcm}(6r, 2h) = rh = 240. \quad (78)$$

## Theorem 5 — the root transition atlas and the positive-edge moment

The directed even-flips of  $C^+$  are the complete graph  $K_{16}$ :

$$\boxed{\begin{aligned} |R^+(E_8)| &= \binom{\dim S^+}{2} = \binom{16}{2} = 120 = \text{Tr}_{S^+} X^2 = 5! = \sum_{m \in (\mathbb{Z}/30\mathbb{Z})^\times} m, \\ |R(E_8)| &= 2 \binom{16}{2} = 240 \end{aligned}} \quad \text{[I]}. \quad (79)$$

( $X = 6Y$ .) So the positive  $E_8$  roots are the unordered transitions of one family; the second hypercharge moment, the positive-root count and the sum of the  $E_8$  exponents coincide at 120. The CP pairing  $m \leftrightarrow 30 - m$  of the exponents gives four pairs, each of sum  $h = 30$ :

$$\boxed{\text{Tr}_{S^+} X^2 = (N_{\text{fam}} + N_\Phi) h = 4 \cdot 30 = 120 = \dim \mathfrak{so}_{16} = \underbrace{45}_{D_5} + \underbrace{15}_{A_3} + \underbrace{60}_{(10,6)}} \quad \text{[I]}, \quad (80)$$

so the hypercharge moment is simultaneously four CP-paired Coxeter periods *and* the adjoint half  $\mathfrak{so}_{16} = (45, 1) \oplus (1, 15) \oplus (10, 6)$  of the  $E_8$  closure.

### How the root *origin* is now obtained — see Theorem 7

A naive flip-size-preserving map  $R_{\text{flip}} \rightarrow \mathbb{R}^8$  cannot be  $E_8$ : the flip split  $240 = 160 + 80$  ( $|F| = 2, 4$ ) differs from the  $D_8$  split  $112 + 128$ . The correct route is *not* a direct 240-metrization but the standard branching  $E_8 \supset D_5 \times A_3$  (Theorem 7), which builds the root spaces from known  $D_5$  and  $A_3$  data. Until the  $A_3$  lattice is derived from the four punctures (Theorem 7, [A]),  $E_8$  remains a Coxeter atlas with a concrete, falsifiable lift.

## Theorem 6 — downstream readouts from $(h, r, c_3)$

All discrete budgets and the seam seed are functions of  $(h, r, c_3) = (30, 8, 1/(8\pi))$ :

$$\dim S^+ = 2r = 16, \quad h = 2(\dim S^+ - 1) = 30, \quad (81)$$

$$10b_1 = 5r + 1 = 41 = 1 + h \frac{\dim S^+}{\dim \mathfrak{g}_{\text{SM}}}, \quad I_1^{\text{EW}} = \frac{5r + 1}{6r} = \frac{41}{48}, \quad (82)$$

$$u = \frac{r}{6} c_3 + 6r c_3^4 = \frac{4}{3} c_3 + 48 c_3^4, \quad \delta_2 = 4! \binom{16}{2} c_3^8 = 2880 c_3^8, \quad (83)$$

$$\xi_{\text{tree}} = \frac{6}{r} = \frac{\dim \mathfrak{g}_{\text{SM}}}{\dim S^+} = \frac{3}{4}, \quad \frac{M_{\text{scal}}^2}{M_{\text{Pl}}^2} = c_3^{\Omega_{\text{adm}} - 10b_1} = c_3^{r-1} = c_3^7. \quad (84)$$

**Two-graph product table.** With the carrier-pair graph  $K_5$  ( $|E(K_5)| = \binom{5}{2} = 10 = \mathcal{A}_\Lambda$ ) and the family-corner graph  $K_4$  on  $\mu_4$  ( $|E(K_4)| = \binom{4}{2} = 6$ ,  $|R(A_3)| = 2|E(K_4)| = 12$ ), every large budget is a product of the two graphs:

$$\boxed{40 = 10 \cdot 4, \quad 60 = 10 \cdot 6, \quad 120 = 10 \cdot 12, \quad 240 = 10 \cdot 24} \quad \text{[I]}, \quad (85)$$

i.e.  $(\sum L, D_{\text{start}}, |R^+(E_8)|, |R(E_8)|) = 10 \cdot (|\mu_4|, |E(K_4)|, |R(A_3)|, |S_4|)$ . Two occupancy identities follow,

$$\boxed{10b_1 = 1 + |\mu_4| |E(K_5)| = 1 + 4 \cdot 10 = 41}, \quad (86)$$

$$\boxed{\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ = (N_{\text{fam}} + N_\Phi) |R(A_3)| = 3 \cdot 16 = 4 \cdot 12 = 48} \quad \text{[I]},$$

so the Higgs singleton is the fourth ( $\mu_4$ ) direction that closes  $N_{\text{fam}} = 3$  to **4** of  $A_3$ . **Occupancy deficit ladder.** The admissible occupancy splits cleanly,

$$\boxed{\begin{aligned} \Omega_{\text{adm}} &= \underbrace{40}_{\sum L(\text{flavor})} + \underbrace{1}_{\text{Higgs}} + \underbrace{7}_{\text{scalaron}} = 48, & \Omega_{\text{adm}} - \sum L &= 8 = h(D_5) = r(E_8), \\ \Omega_{\text{adm}} - 10b_1 &= 7 = h(D_5) - 1 \end{aligned}} \quad \text{[I]}, \quad (87)$$

so the scalaron exponent  $r - 1 = 7$  is exactly the post-flavor, post-Higgs occupancy remainder,  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^{\Omega_{\text{adm}} - 10b_1} = c_3^7$ . With  $h(D_5) = 8$  and  $|R^+(A_3)| = 6$  the seed itself is a  $D_5/A_3$  ratio-product,  $u = \frac{h(D_5)}{|R^+(A_3)|} c_3 + h(D_5) |R^+(A_3)| c_3^4$ , and  $\xi = \frac{|R^+(A_3)|}{h(D_5)} (1 + |R^+(A_3)|^2 c_3^3)^{-1} = \frac{3}{4} (1 + 36c_3^3)^{-1}$ . The seed readouts and the two decuples are

$$\lambda_C^2 = u(1-u), \quad \beta_{\text{rad}} = 2c_3u, \quad \sin^2 \theta_{13} = e^{-5/6}u, \quad \boxed{\omega_b = \frac{\lambda_C}{\mathcal{A}_\Lambda} = \frac{\lambda_C}{10} = 0.02244} \quad (\text{Planck } 0.02237), \quad (88)$$

$$\boxed{\begin{aligned} \det M_\ell &\propto u^{h/N_{\text{fam}}} = u^{10}, \\ \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} &= \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} \left[ \frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}} \right]^{h/N_{\text{fam}}}, \end{aligned}} \quad (89)$$

with lepton powers  $(5, 3, 2) = (g_{\text{car}}, b, s)$ :  $\sum = 10 = \mathcal{A}_\Lambda$  (the  $\Lambda$  decuple),  $\prod = 30 = h$  (the Coxeter period). The first-generation transport sums read the non-primitive  $\mathbb{Z}_{30}$  load,

$$(G_1, G_2, G_3) = (22, 13, 5) = (h - r, \dim \mathfrak{g}_{\text{SM}} + 1, g_{\text{car}}), \quad h - r = 30 - \varphi(30) = 22, \quad (90)$$

**Flavor transport is a  $D_5 \times A_3$  object.** The transport packets  $L_u=(7, 3, 0)$ ,  $L_d=(7, 5, 2)$ ,  $L_e=(8, 5, 3)$  split into a  $D_5$  budget and an  $A_3$  geometry:

$$\boxed{\sum_{f,j} L_{f,j} = 40 = |R(D_5)| \text{ (Clebsch edges)}, \quad L_{f,j} \bmod 6 \text{ lives on } C_6 = Z_6 = |R^+(A_3)|} \quad \text{[I]}, \quad (91)$$

the residues being the  $D_6$ -orbit of the perfect set  $\{0, 1, 3\} \subset C_6$  (cyclic distances  $\{1, 2, 3\}$ ):  $u \equiv \{0, 1, 3\}$ ,  $d \equiv 2 - \{0, 1, 3\}$ ,  $e \equiv 2 + \{0, 1, 3\}$ , with the universal first-generation +6 winding. So the carrier  $D_5$  sets the total budget and the family  $A_3$  (its 6-element positive-root hexagon) sets the per-entry distances. The scalaron mass is a pure seam power [I]; the amplitude is a *conditional* readout, kept [P]/[A] until the scalaron variable is formally identified with the Starobinsky  $M$  parameter. That identification is now *heat-kernel grounded* (G2): the spectral-action Seeley–DeWitt coefficient  $a_4|_{R^2} \sim R^2/72$  delivers exactly the Starobinsky  $R^2$  term, with  $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$  and the TFPT closure  $c_3^7$  fixing the cutoff moment  $f_0$  ([verification/v36\_spectral\_action\_g2.py]):

$$\underbrace{\frac{M_{\text{scal}}^2}{\bar{M}_{\text{Pl}}^2} = c_3^{r-1} = c_3^7}_{\text{[I]}}, \quad \underbrace{A_s = \frac{N_\star^2}{24\pi^2} \frac{M_{\text{scal}}^2}{\bar{M}_{\text{Pl}}^2} = \frac{N_\star^2 c_3^7}{24\pi^2} = 2.17 \times 10^{-9}}_{\text{[P]/[A] (if } M_{\text{scal}} = M_{\text{Star}})}, \quad (92)$$

## Theorem 7 — the $D_5 \times A_3$ lift of the $E_8$ atlas

Since  $A_3 = \mathfrak{su}_4 \cong \mathfrak{so}_6 = D_3$ , one has the standard chain  $E_8 \supset \mathfrak{so}_{16} \supset \mathfrak{so}_{10} \oplus \mathfrak{so}_6 = D_5 \times A_3$ , with the exact branching (verified dimensions and root/weight counts):

$$\boxed{\mathfrak{e}_8 = (45, 1) \oplus (1, 15) \oplus (10, 6) \oplus (16, 4) \oplus (\overline{16}, \overline{4})}, \quad (93)$$

$$\underbrace{45+15+60+64+64}_{=248} = \dim E_8, \quad \underbrace{40+12+60+64+64}_{=240} = |R(E_8)|.$$

The adjoint **120** of  $\mathfrak{so}_{16}$  is  $(45, 1) \oplus (1, 15) \oplus (10, 6)$  and its spinor **128** is  $(16, 4) \oplus (\overline{16}, \overline{4})$ . The TFPT dictionary makes each summand a known object:

| piece                                          | value         | TFPT reading                                                                                           |
|------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------|
| $(45, 1)$                                      | $45 = 40 + 5$ | $D_5$ carrier adjoint: Clebsch edges $40 =  R(D_5) $ plus carrier rank $g_{\text{car}} = 5$            |
| $(1, 15)$                                      | $15 = 12 + 3$ | $A_3$ family+Higgs adjoint: $\dim \mathfrak{g}_{\text{SM}} = 12$ plus family rank $N_{\text{fam}} = 3$ |
| $(10, 6)$                                      | $10 \cdot 6$  | $\Lambda$ decuple $\mathcal{A}_\Lambda = \binom{5}{2} = 10$ times $Z_6$ transport $\binom{4}{2} = 6$   |
| $(16, 4) \oplus (\overline{16}, \overline{4})$ | $2 \cdot 64$  | matter half-spinor $S^+$ times the four corners $\mu_4$ , plus conjugate sheet                         |

**Proof route (replaces the old blind 240-metrization).** *Stage 1* ( $D_5$ , [I]):  $C^+$  is the  $D_5$  half-spin orbit,  $R_4 = \text{Clebsch}$ ,  $\text{Aut} = W(D_5)$ ,  $|E(R_4)| = 40 = |R(D_5)|$  — Theorem 2. *Stage 2* ( $A_3$ , [I]): Paper 2 fixes  $X_f^\circ = \mathbb{P}^1 \setminus \mu_4$  with  $H_1(X_f^\circ, \mathbb{Z}) = \mathbb{Z}^3$  and a rigid  $D_4 \subset PGL_2$  generated by  $z \mapsto iz$  and  $z \mapsto 1/z$ . The logarithmic forms  $\omega_i = dz/(z - p_i)$  regular at infinity span  $H^1$  by their differences, dual under the residue pairing to the puncture loops  $\gamma_i$  with  $\text{Res}(\omega_i, \gamma_j) = \delta_{ij}$ ; hence the canonical (residue) metric on the puncture classes is  $I_4$ , and the twelve puncture-difference cycles  $\gamma_i - \gamma_j$  are exactly the  $A_3$  roots:

$$\boxed{\begin{aligned} \{\gamma_i - \gamma_j\}_{i \neq j}: \# = 12 = |R(A_3)| = \dim \mathfrak{g}_{\text{SM}}, \quad \text{rank} = 3 = N_{\text{fam}}, \\ \text{Gram} = \text{Cartan}(A_3), \det = 4 \end{aligned}} \quad \text{[I]}, \quad (94)$$

with  $\dim A_3 = 15 = \dim S^+ - 1$ , and the  $A_3$  positive roots equal the exponent sum  $1+2+3 = 6 = Z_6 = |E(K_4)|$ . *The corner rotation realises the  $A_3$  Coxeter element.* The Paper-2 generator  $z \mapsto iz$  permutes  $\mu_4$  as the 4-cycle  $(p_1 p_2 p_3 p_4) \in S_4 = W(A_3)$  and acts on  $H^1 = \mathbb{C}^3$  as an isometry of the Cartan form with

$$\boxed{\text{ord} = 4 = h(A_3) = |\mu_4|, \quad \text{spec} = \{i, -1, -i\}, \quad \chi(t) = \frac{t^4 - 1}{t - 1} = \Phi_4(t)\Phi_2(t)} \quad \text{[I]}, \quad (95)$$

i.e. the eigenvalues are the  $A_3$  exponents  $\{1, 2, 3\}$  as fourth roots of unity — the exact analogue of the  $E_8$  Coxeter element reading  $\Phi_{30}$ . The full  $D_4 = \langle z \mapsto iz, z \mapsto 1/z \rangle$  embeds in  $W(A_3) = S_4$  (order 8, index 3) by isometries. *Stage 3 (standard, [I]):* assemble  $E_8$  from the  $D_5 \times A_3$  branching — no 240-map required.

**The  $\mu_4$  glue theorem (why the four corners appear everywhere).** The two factors have equal discriminant groups, and  $E_8$  is unimodular:

$$\boxed{\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4, \quad [E_8 : D_5 \oplus A_3] = \sqrt{\frac{\det D_5 \cdot \det A_3}{\det E_8}} = \sqrt{\frac{4 \cdot 4}{1}} = 4 = |\mu_4|} \quad \text{[I]}. \quad (96)$$

So  $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$ : the four family corners  $\mu_4 = \mathbb{Z}_4$  are precisely the glue code that fuses the carrier and the family into the unimodular  $E_8$  lattice. The glue pairs the  $D_5$  spinor class ( $= S^+$ , the family) with the  $A_3$  class — exactly the  $(16, 4) \oplus (\overline{16}, \overline{4})$  spinor block. Hence  $\mu_4$  plays three roles at once: family corners,  $A_3$  origin, and  $E_8$  glue.

**Discriminant-norm glue lemma (and the TFPT 5/4, 3/4 identification).** The two glue generators carry the discriminant quadratic form  $q \in \mathbb{Q}/2\mathbb{Z}$ :

$$\boxed{\begin{aligned} q(D_5) &= \frac{n}{4} \Big|_{n=5} = \frac{5}{4} \text{ (spinor class)}, & q(A_3) &= \frac{n}{n+1} \Big|_{n=3} = \frac{3}{4} \text{ (corner class)}, \\ q(D_5) + q(A_3) &= 2 \equiv 0 \pmod{2}, & q(D_5) - q(A_3) &= \frac{1}{2} \end{aligned}} \quad \text{[I]}. \quad (97)$$

The diagonal  $\langle(s, g)\rangle \cong \mathbb{Z}_4$  is isotropic ( $k^2 \cdot 2 \equiv 0$  for all  $k$ ), so the glue closes to an *even unimodular* lattice — this is  $E_8$ .

**The two glue norms are just  $(g_{\text{car}}, N_{\text{fam}})$  over the glue index.** Because  $\text{rank } D_5 = g_{\text{car}} = 5$  (so  $q(D_5) = \frac{n}{4}|_{n=5} = g_{\text{car}}/|\mu_4|$ ) and  $\text{rank } A_3 = N_{\text{fam}} = 3$  (so  $q(A_3) = \frac{n}{n+1}|_{n=3} = N_{\text{fam}}/|\mu_4|$ , with  $n+1 = 4 = |\mu_4|$ ),

$$\boxed{q(D_5) = \frac{g_{\text{car}}}{|\mu_4|} = \frac{5}{4}, \quad q(A_3) = \frac{N_{\text{fam}}}{|\mu_4|} = \frac{3}{4}}$$

and *all four arithmetic operations* on the pair are then forced compiler atoms:

$$\begin{aligned} \underbrace{q(D_5) + q(A_3) = \frac{g_{\text{car}} + N_{\text{fam}}}{|\mu_4|} = 2}_{E_8 \text{ root norm}}, & \quad \underbrace{q(D_5) - q(A_3) = \frac{|\mathbb{Z}_2|}{|\mu_4|} = \frac{1}{2}}_{\delta \text{ (lepton transport)}}, \\ \underbrace{q(D_5)q(A_3) = \frac{g_{\text{car}}N_{\text{fam}}}{|\mu_4|^2} = \frac{15}{16}}_{\dim \mathfrak{su}(4)/\dim S^+}, & \quad \underbrace{\frac{q(D_5)}{q(A_3)} = \frac{g_{\text{car}}}{N_{\text{fam}}} = \frac{5}{3}}. \end{aligned}$$

So the distinguished lepton transport value  $\delta = \frac{1}{2} = (g_{\text{car}} - N_{\text{fam}})/|\mu_4| = |\mathbb{Z}_2|/|\mu_4|$  is the carrier-minus-family count over the glue index — a fully atomic origin, not a chosen rational (it also equals the harmonic-frame holonomy diagonal modulus, `[verification/v41_leg_assignment.py]`); and (sum, ratio) invert uniquely back to  $(\frac{5}{4}, \frac{3}{4})$ . `[verification/v51_boundary_half_step.py]` The sum  $\frac{5}{4} + \frac{3}{4} = 2$  is literally the  $E_8$  spinor-root norm  $\frac{1}{2}(\pm)^8$  split across  $D_5 \times D_3$ :  $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 2$ . Strikingly, both norms are pre-existing TFPT constants:

$$\boxed{\frac{\delta_2}{\delta_{\text{top}}^2} = \frac{2880 c_3^8}{(48 c_3^4)^2} = \frac{5}{4} = q(D_5), \quad \xi_{\text{tree}} = \frac{\dim \mathfrak{g}_{\text{SM}}}{\dim S^+} = \frac{12}{16} = \frac{3}{4} = q(A_3)} \quad \textcolor{green}{\mathbb{I}}, \quad (98)$$

so  $\delta_2/\delta_{\text{top}}^2 + \xi_{\text{tree}} = 2$ : the two-loop/defect ratio is the  $D_5$  spinor norm and the gravitational tree normalization is the  $A_3$  corner norm, and their sum is the  $E_8$  root norm.

**The whole integer skeleton from one pair  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$**  \mathbb{I}  
(`[verification/v53_compiler_core.py]`)

Every load-bearing integer of the compiler flows from the single boundary pair  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$  by sum, difference and squares:

$$\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = 8, \quad |\mathbb{Z}_2| = g_{\text{car}} - N_{\text{fam}} = 2, \quad |\mu_4| = \frac{g_{\text{car}} + N_{\text{fam}}}{2} = 4,$$

and  $(N_{\text{fam}}, |\mu_4|, g_{\text{car}}) = (3, 4, 5)$  is *the* Pythagorean triple. This is not decorative: the grand-mass-volume exponent  $\Delta_Y = g_{\text{car}}^2 = 25$  ( $\det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{25}$ , §Paper 2) splits over the sectors as a *difference of squares*,

$$\boxed{\Delta_Y = g_{\text{car}}^2 = \underbrace{N_{\text{fam}}^2}_{\text{down}=9} + \underbrace{(g_{\text{car}} - N_{\text{fam}})(g_{\text{car}} + N_{\text{fam}})}_{=|\mathbb{Z}_2| \cdot \text{rank } E_8} = 9 + 16 = \dim S^+}$$

i.e. the down sector carries  $N_{\text{fam}}^2$  and up+lepton carries  $|\mathbb{Z}_2| \cdot \text{rank } E_8 = \dim S^+$  (the  $K$ -row sums  $(6, 9, 10)$ ). **The anchor is the Dirac square root of this skeleton:**  $a = (1, 1, 2)$  is the *unique* 3-multiset whose elementary symmetric functions are  $(|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|) = (4, 5, 2)$ , equivalently its characteristic polynomial is  $\chi_a(t) = (t-1)^2(t-2) = t^3 - |\mu_4|t^2 + g_{\text{car}}t - |\mathbb{Z}_2|$  — the compiler atoms *are* its char-poly coefficients. So the theory's only transcendental is  $\pi$  (through  $c_3 = \frac{1}{8\pi}$ ,  $8 = \text{rank } E_8$ , and the seed  $\frac{1}{6\pi}$ ,  $6 = |R^+(A_3)|$ ); everything else is this  $(5, 3)$ -generated integer skeleton.



## Boundary Carrier Selection Theorem (Theorem A) /[P]

$E_8$  is not assumed: it is *selected* as the unique even-unimodular audit hull of the admissible boundary-carrier data. **Boundary side:**  $\mu_4 = \{1, i, -1, -i\}$  gives  $X_f = \mathbb{P}^1 \setminus \mu_4$  with  $H_1 = \mathbb{Z}^3$ , hence  $A_3$  and  $N_{\text{fam}} = 3$ . **Carrier side:** a minimal  $D$ -type carrier with a 16-dim half-spinor forces  $D_5$  (only  $n = 5$  gives  $2^{n-1} = 16$ ;  $S^+ = \Lambda^{\text{even}} \mathbb{C}^5 = 1+10+5$ ). **Glue:**  $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$  (isomorphic, cyclic, anti-isometric),  $q(D_5) + q(A_3) = 2$  (the  $E_8$  root norm), glue index  $4 = |\mu_4|$ , and  $E_8$  is the unique even unimodular rank-8 lattice. **Exclusion:** the four conditions

$$(i) \dim S^+ = 16, \quad (ii) N_{\text{fam}} = 3, \quad (iii) \text{cyclic } \mathbb{Z}_4 \text{ glue}, \quad (iv) q_D + q_A = 2$$

are met *only* by  $D_5 \oplus A_3$ ;  $D_8$  (non-cyclic  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , no family geometry),  $E_7 \oplus A_1$  (1 family, no 16-spinor),  $E_6 \oplus A_2$  (2 families,  $\mathbb{Z}_3$ ),  $A_4 \oplus A_4$  (no  $D$ -spinor,  $\mathbb{Z}_5$ ) and  $A_8$  (single factor) each fail at least one. **Typing:** the lattice/classification facts are [L]; the physical selection (reflection-positive seam  $c_3 = 1/8\pi$  + minimal chiral carrier  $\Rightarrow$  exactly these inputs) is the [P]/[A] selection axiom (P1 analytic interface + P2 carrier interface), not a zero-input derivation. [verification/v47\_selection\_theorem.py]

**Group-level closure ( $\mu_4$  as the common centre).** The lattice glue lifts to a clean group statement, because both factors share the *same* centre and  $\mu_4$  *is* the diagonal:

$$Z(\text{Spin}(10)) = \mathbb{Z}_4, \quad Z(SU(4)) = \mathbb{Z}_4, \quad |\mu_4| = 4, \quad G_{\text{closure}} = \frac{\text{Spin}(10) \times SU(4)}{\Delta \mathbb{Z}_4} \subset E_8 \quad [\text{I}], \quad (99)$$

with  $\text{rank} = 5 + 3 = 8 = \text{rank } E_8$ . Every adjoint block is diagonal- $\mathbb{Z}_4$  neutral (generator  $\omega = i$ ; charges mod 4):  $(45, 1), (1, 15) : 0$ ;  $(10, 6) : 2+2=0$ ;  $(16, 4) : 1+3=0$ ;  $(\overline{16}, \overline{4}) : 3+1=0$ , so  $45+15+60+64+64 = 248$  are genuine reps of the quotient. This promotes  $\mu_4$  from a lattice glue index to the *common centre quotient* — the physically cleanest language for the gluing: the  $D_5$  half-spinor 16 pairs with the  $SU(4)$  fundamental 4 so the spinor charges  $\pm 1$  cancel the fundamental charges  $\mp 1$ .

**Root-length completion.** Every  $E_8$  block reaches norm 2:

| block                                          | norm split                      | reading                               |
|------------------------------------------------|---------------------------------|---------------------------------------|
| $(45, 1), (1, 15)$                             | 2, 2                            | $D_5, A_3$ root adjoints              |
| $(10, 6)$                                      | $1 + 1 = 2$                     | $SO(10)$ vector $\oplus SO(6)$ vector |
| $(16, 4) \oplus (\overline{16}, \overline{4})$ | $\frac{5}{4} + \frac{3}{4} = 2$ | $D_5$ spinor norm + $A_3$ corner norm |

The old question “can the 240 flip atlas be metrized to  $E_8$ ?” is now the sharp question “are the TFPT  $D_5$  and  $A_3$  metrics in standard normalization?” — and the  $5/4, 3/4$  match says yes.

**Explicit construction (the lift is now built, not counted).** Embedding  $D_5 \oplus A_3 = D_5 \oplus D_3 \subset D_8$  and adjoining the even spinor coset (the  $\mu_4$  glue), the 240 roots are produced explicitly and verified:

$$\{\pm e_i \pm e_j\}_{i < j}^8 (112) \cup \{\frac{1}{2}(\pm)^8 : \text{even}\} (128) = 240 \text{ roots, all norm 2, Cartan det} = 1 \quad [\text{L}]. \quad (100)$$

Splitting each root into its  $D_5$  (first five) and  $A_3$  (last three) parts gives exactly the branching blocks —  $(2, 0) : 40$ ,  $(0, 2) : 12$ ,  $(1, 1) : 60$ , and  $(\frac{5}{4}, \frac{3}{4}) : 128$  — and the 128 spinor roots realise  $\frac{5}{4} + \frac{3}{4} = 2$  on the nose. All 240 vectors are integer combinations of one set of 8 simple roots, so the assembled lattice *is*  $E_8$ . The blind 240-metrization is therefore eliminated:  $E_8$  is explicitly  $(D_5 \oplus A_3) + \mu_4$ -glue.

**Remark** (the metric, computed). This reframes  $E_8$  from origin to closure:  $E_8$  is the closure algebra of the  $D_5$  carrier and the  $A_3$  family geometry, glued by  $\mu_4$ . The root system, Cartan



form,  $W(A_3) = S_4$  and the corner-rotation =  $A_3$  Coxeter element are exact (residue pairing + explicit Paper-2 symmetry). The transcendental geometric metric  $G_{ij} = -\log |p_i - p_j|$  on  $\mu_4$  has been computed: it is only  $D_4$ -symmetric, with a regularisation-independent anisotropy  $-\log 2$  on the root space, hence *not* proportional to  $\text{Cartan}(A_3)$ . It need not be: the  $E_8$ -relevant  $A_3$  is the canonical *residue* pairing, and the geometric metric is a  $D_4$ -equivariant deformation that preserves the root system, Weyl group and Coxeter element. So the lattice data is exact [I]; the analytic metric is a *characterised* deformation, not an open gap.

**Uniform Coxeter law.** All three factors obey  $|R| = h \cdot \text{rank}$ , and the two TFPT Coxeter numbers are themselves carrier data:

$$\boxed{\begin{aligned} h(D_5) &= 8 = r(E_8) = \varphi(30), & h(A_3) &= 4 = |\mu_4|, \\ h(D_5)h(A_3) &= 2^{g_{\text{car}}} = 32, & 2^{g_{\text{car}}} - 2 &= 30 = h(E_8) \end{aligned}} \quad [\mathbf{I}], \quad (101)$$

so the Coxeter closure of Theorem 3 reappears as  $h(E_8) = h(D_5)h(A_3) - 2$ , and  $h(D_5) + h(A_3) = 12 = \dim \mathfrak{g}_{\text{SM}} = |R(A_3)|$ .

## Theorem 8 — the $(1, 1, 2)$ anchor microcode and the $\varphi_0$ -ladder

The parabolic flavor bundle  $E_\bullet = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  on  $\mathbb{P}^1 \setminus \mu_4$  has splitting type  $a = (1, 1, 2)$ , whose power sums  $p_n(a) = 2 + 2^n$  generate the whole compiler budget:

### The anchor power compiler

$$(p_1, \dots, p_5) = (4, 6, 10, 18, 34) = (|\mu_4|, |R^+(A_3)|, \mathcal{A}_\Lambda, N_{\text{fam}}|R^+(A_3)|, \mathbb{Z}_6 \text{ lift}),$$

$$q(A_3) = \frac{p_2}{2p_1} = \frac{3}{4}, \quad q(D_5) = \frac{p_3}{2p_1} = \frac{5}{4}, \quad \sum L = p_1 p_3 = 40,$$

$$\Omega_{\text{adm}} = 2p_1 p_2 = 48, \quad D_{\text{start}} = p_2 p_3 = 60, \quad |R(E_8)| = 4p_2 p_3 = 240.$$

The elementary symmetric data  $\chi_a(t) = (t-1)^2(t-2)$  give  $(e_1, e_2, e_3) = (4, 5, 2) = (|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|)$ , so the anchor reconstructs the carrier rank, and  $10b_1 = e_1^2 + e_2^2 = 4^2 + 5^2 = 41$ . The seed itself is the two-term anchor form  $\varphi_0^{\text{ret}} = \frac{2p_1}{p_2}c_3 + 2p_1 p_2 c_3^4 = \frac{4}{3}c_3 + 48c_3^4$ .

**The  $\varphi_0$ -ladder.** With  $\lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}$ , every fermion mass is the single closed form  $\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}$  with  $\lambda_Y^L \Lambda = \pi c (\varphi_0^{\text{ret}})^k$ , so all nine holonomies are resolvent outputs and the masses form the  $\varphi_0^{\text{ret}}$ -order ladder  $t(0) > \{b, c, \tau\}(2) > \{s, \mu\}(3) > \{u, d\}(4) > e(5)$ .

**Secondary  $E_8$  atlas.** Beyond the main  $D_5 \times A_3$  split, the residue matrix appears in the  $E_6 \times A_2$  branching,  $248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$ , and the scalaron exponent in  $E_7 \times A_1$ ,  $112 = 7 \cdot 16$ . The seven  $E_8$  slices are charted in the companion note `e8_secondary_branching_atlas` as a falsifiable audit raster.

## Master statement and status

### One line

Five carrier slots give  $D_5$ , four family corners give  $A_3$ , and  $D_5 \times A_3$  gives the  $E_8$  atlas.

Equivalently:  $\mathcal{A}_\Lambda = \frac{h}{N_{\text{fam}}} = 10 = \binom{5}{2}$ ,  $h = 30 = 2 \cdot 3 \cdot 5$ ,  $(5, 3, 2) : \sum = 10, \prod = 30$ ;  $\Lambda$ ,  $\det M_\ell$ , the  $E_8$  Coxeter structure and the action ladder read the same five-slot pair-decuple. The Standard model, flavor,  $\Lambda$ , inflation and  $E_8$  are projections of the even carrier code  $C^+$  ( $= D_5$  half-spinor) on five slots together with the four-point family  $A_3$ .

| Statement                                                                                                      | Status  | Note                                   |
|----------------------------------------------------------------------------------------------------------------|---------|----------------------------------------|
| Even code $C^+$ , $\dim S^+ = 16$ , dual-root $Y$                                                              | [I]     | Theorem 1                              |
| Exterior–Action duality $(1, 10, 5) \leftrightarrow (1, 5, 10)$                                                | [I]     | scheme eigenvalues verified            |
| $R_4 = \text{Clebsch } (16, 5, 0, 2)$ , $R_2 = (16, 10, 6, 6)$ ; $\text{Aut}_{\text{aff}} = W(D_5) = 1920$     | [I]     | carrier = $D_5$ half-spinor            |
| $2^g - 2 = g(g+1) \Leftrightarrow g = 5$                                                                       | [I]     | Coxeter closure (unique)               |
| $h=30=2\cdot 3\cdot 5$ , $r=\varphi(30)=8$ , $\chi_C=\Phi_{30}$                                                | [I]/[P] | cyclotomic compiler                    |
| $ R^+  = \binom{16}{2} = \text{Tr } X^2 = 4h = 120$                                                            | [I]     | positive-edge moment                   |
| $40, 60, 120, 240 = 10 \cdot (4, 6, 12, 24)$ ; $41=1+4\cdot 10$                                                | [I]     | $K_5 \times K_4$ product table         |
| $10b_1=5r+1$ , $u$ totient, $M_{\text{scal}}^2=c_3^{r-1}$ , $\omega_b=\lambda_C/10$                            | [I]/[P] | downstream readouts                    |
| $A_s = N_*^2 c_3^7 / 24\pi^2$                                                                                  | [P]/[A] | if $M_{\text{scal}} = M_{\text{Star}}$ |
| $\mathfrak{e}_8 = (45, 1) \oplus (1, 15) \oplus (10, 6) \oplus (16, 4) \oplus (\overline{16}, \overline{4})$   | [I]/[P] | $D_5 \times A_3$ lift                  |
| 240 roots built from $D_5 \oplus A_3 + \mu_4$ glue, det 1                                                      | [L]     | explicit $E_8$ construction            |
| flavor $\sum L=40= R(D_5) $ , entries on $C_6= R^+(A_3) $                                                      | [I]     | transport = $D_5 \times A_3$           |
| $A_3$ from $\mathbb{P}^1 \setminus \mu_4$ : residue pairing = Cartan, det 4                                    | [I]     | Stage 2 (Theorem 7)                    |
| corner $z \mapsto iz = A_3$ Coxeter el.: $\{i, -1, -i\}$ , $\Phi_4 \Phi_2$                                     | [I]     | order 4 = $h(A_3) =  \mu_4 $           |
| $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ , glue = $\mu_4$                                          | [I]     | $E_8 = (D_5 \oplus A_3) + \mu_4$       |
| $q(D_5) = \frac{5}{4} = \delta_2 / \delta_{\text{top}}^2$ , $q(A_3) = \frac{3}{4} = \xi_{\text{tree}}$ , sum 2 | [I]     | discriminant-norm glue                 |
| $120 = \dim \mathfrak{so}_{16}$ ; $\Omega_{\text{adm}} = 40 + 1 + 7$                                           | [I]     | adjoint half; deficit ladder           |
| $h(D_5) = 8 = r(E_8)$ , $h(A_3) = 4 =  \mu_4 $ , $ R  = h \cdot r$                                             | [I]     | uniform Coxeter law                    |
| $\text{Exp}(E_8)$ as 41/48 budget deficits                                                                     | [A]     | exponent ledger (fingerprint)          |
| geometric metric $D_4$ -deform (anisotropy – log 2), residue = $A_3$                                           | [I]     | metric computed/characterised          |

*Scope.* This note compresses; it does not re-derive the upstream boundary kernel (Paper 1) or the carrier rigidity theorem (Paper 2), and it makes no claim of a final theory. The arithmetic and the  $D_5/\text{Clebsch}/W(D_5)$ ,  $A_3/\text{residue-Cartan}/\text{Coxeter}$ , and  $\mu_4$ -glue identifications are exact, and the  $E_8$  lattice is now built explicitly from  $D_5 \oplus A_3$  plus the  $\mu_4$  glue (240 roots, det 1). The only residual [P] is the analytic identification of Paper 2’s  $L^2$  cusp Hodge metric with the canonical residue metric up to scale, which does not affect the explicit root/lattice/Weyl construction. Honest non-fits (Koide  $0.6645 \neq 2/3$ ,  $\eta_B$ ,  $m_p/m_e$ ) stay outside the compiler.

## Appendix: machine-checkable $E_8 = (D_5 \oplus A_3) + \mu_4$ construction

The lift is reproducible by a short script (all entries below verified exactly).

### Cartan matrices and discriminant groups.

$$C(D_5) = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}, \quad C(A_3) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}; \quad \det C(D_5) = \det C(A_3) = 4.$$

Smith normal forms give  $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ ; the glue index is  $[E_8 : D_5 \oplus A_3] = \sqrt{4 \cdot 4/1} = 4 = |\mu_4|$ , with discriminant-form norms  $q(D_5) = \frac{5}{4}$ ,  $q(A_3) = \frac{3}{4}$ ,  $q(D_5) + q(A_3) = 2 \equiv 0 \pmod{2}$  (even-glue). Writing  $\mu := |\mu_4| = 4$  the norms are  $q(D_5) = \frac{\mu+1}{\mu}$ ,  $q(A_3) = \frac{\mu-1}{\mu}$ , so  $q(D_5)q(A_3) = 1 - \frac{1}{\mu^2} = \frac{15}{16}$  and

$$\dim S^+(1 - q(D_5)q(A_3)) = 16 \cdot \frac{1}{16} = 1 \quad [\text{I}], \quad (102)$$

i.e. the glue product fills all  $15 = \dim S^+ - 1$  non-trivial half-spin states and the defect is exactly the singleton (the  $\nu^c/\text{Higgs}$  closure). And  $E_8$  has a tidy  $\mu_4$ -cardinal form,  $\dim E_8 = \binom{|\mu_4|^2}{2} + 2|\mu_4|^3 = \binom{16}{2} + 2 \cdot 64 = 120 + 128 = 248$  [P].

**Explicit roots.** Split  $\mathbb{R}^8 = \mathbb{R}_{(D_5)}^5 \oplus \mathbb{R}_{(A_3)}^3$ . The 240 roots are

$$\{\pm e_i \pm e_j : 1 \leq i < j \leq 8\} (112) \cup \{\tfrac{1}{2}(\pm 1)^8 : \text{even \# of } -\} (128),$$

all of norm 2. Splitting each by  $(\|\cdot\|_{D_5}^2, \|\cdot\|_{A_3}^2)$  gives the branching blocks

$$(2, 0): 40 [D_5], \quad (0, 2): 12 [A_3], \quad (1, 1): 60 [(10, 6)], \quad (\tfrac{5}{4}, \tfrac{3}{4}): 128 [(16, 4) \oplus (\overline{16}, \overline{4})],$$

so  $\mathfrak{e}_8 = (45, 1) \oplus (1, 15) \oplus (10, 6) \oplus (16, 4) \oplus (\overline{16}, \overline{4})$ ,  $248 = 45+15+60+64+64$ ,  $240 = 40+12+60+64+64$ .

**Simple-root Gram (unimodular check).** With the Bourbaki simple roots  $\alpha_1 = \frac{1}{2}(e_1 - e_2 - \dots - e_7 + e_8)$ ,  $\alpha_2 = e_1 + e_2$ , and  $\alpha_k = e_{k-1} - e_{k-2}$  for  $k = 3, \dots, 8$  (so  $\alpha_3 = e_2 - e_1$ ,  $\alpha_4 = e_3 - e_2$ ,  $\dots$ ,  $\alpha_8 = e_7 - e_6$ ), the Gram matrix  $G_{ij} = \langle \alpha_i, \alpha_j \rangle$  is the  $E_8$  Cartan matrix with  $\det G = 1$  and all  $G_{ii} = 2$ ; every one of the 240 roots is an integer combination of the  $\alpha_i$ . Hence the assembled lattice is  $E_8$ , not merely a 240-count. [F] (The shifted index  $e_{k-2} - e_{k-3}$  would duplicate  $\alpha_3$  and make  $G$  singular — use  $e_{k-1} - e_{k-2}$ .)

## Part III

# Appendix: ablation of the EM fixed point

The single sharpest reviewer test is: *was the closure  $F_{U(1)}(\alpha) = 0$  engineered so that  $\alpha$  comes out?* The honest answer is an ablation — vary each input and see which actually move the root. Three one-parameter scans (all others held at their compiler values) separate the *load-bearing* integers from the *fine* last-digit selector.

EM fixed point: only  $M = 41$  and  $N = 8$  land on  $\alpha^{-1}$

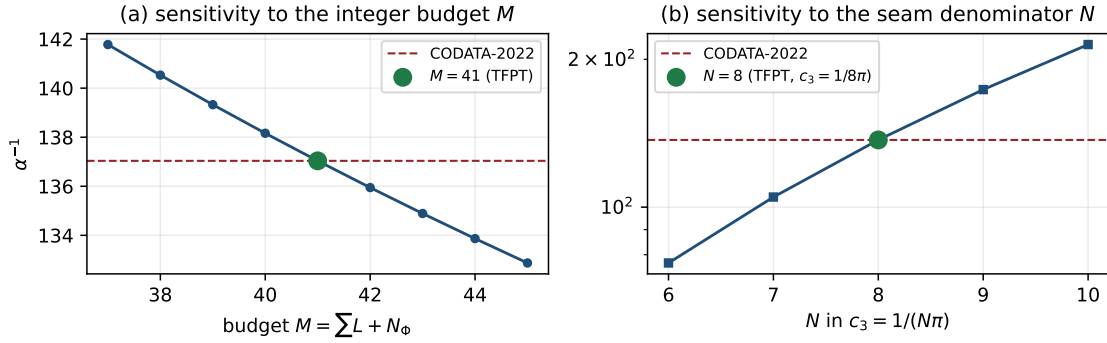


Figure 7: EM-closure ablation (data plot, `verification/make_figures.py`). **(a)**  $\alpha_*^{-1}$  vs the integer budget  $M$ : only  $M = 41$  lands on CODATA-2022. **(b)**  $\alpha_*^{-1}$  vs the seam denominator  $N$  in  $c_3 = 1/(N\pi)$ : only  $N = 8$  lands. Both integers are fixed elsewhere (flavor budget; rank/ $h(D_5)$ ), so the equation is not freely tunable to 137.036.

**(A1) The transport budget  $M = \sum L + N_\Phi$  (exponent held at  $-\frac{5}{4}$ ):**

| $M$             | 38      | 39      | 40      | 41                 | 42      | 43      | 44      |
|-----------------|---------|---------|---------|--------------------|---------|---------|---------|
| $\alpha_*^{-1}$ | 140.530 | 139.326 | 138.162 | <b>137.0359992</b> | 135.946 | 134.890 | 133.866 |

Each unit of  $M$  shifts  $\alpha_*^{-1}$  by  $\approx 1.09$ , so *only*  $M = 41$  lands on 137.036. This is a hard lock — and  $41 = 10b_1 = \sum L + N_\Phi = 40 + 1$  is fixed by the compiler (flavor budget 40 + one Higgs). **Load-bearing.**

**(A2) The seam “8” in  $c_3 = 1/(N\pi)$  (here  $\delta_{\text{top}} = 48c_3^4$  tracks  $c_3$ ):**

| $N$             | 6     | 7      | 8              | 9      | 10     |
|-----------------|-------|--------|----------------|--------|--------|
| $\alpha_*^{-1}$ | 76.97 | 104.85 | <b>137.036</b> | 173.53 | 214.34 |

Only  $N = 8$  — i.e.  $c_3 = 1/(8\pi) = 1/(\text{rank } E_8 \cdot \pi)$  — reproduces 137.036. The “8” is the same one fixed by the bootstrap ( $h(D_5) = \text{rank } E_8 = \varphi(30)$ ) and the Gauss–Bonnet hardening. **Load-bearing.**

**(A3) The carrier discriminant exponent (at  $M = 41$ ,  $c_3 = 1/(8\pi)$ ):**

| exponent        | $-\frac{1}{4}$ | $-\frac{3}{4}$ | $-\frac{5}{4}$     | $-\frac{7}{4}$ | $-\frac{9}{4}$ |
|-----------------|----------------|----------------|--------------------|----------------|----------------|
| $\alpha_*^{-1}$ | 137.035995     | 137.035997     | <b>137.0359992</b> | 137.036001     | 137.036003     |

The exponent moves only the *seventh* digit; all candidates agree to six. Here  $-\frac{5}{4} = q(D_5)$  is selected, but this is a *fine* selector, not a hard lock. **Fine (last-digit).**

### Ablation verdict — honest

The EM fixed point is a **hard test of two integers** — the budget  $M = 41$  and the seam denominator  $8 = \text{rank } E_8$ , each decisive (a single-unit change misses by  $\sim 1$  in  $\alpha_\star^{-1}$ ) — and a **fine test of one norm**, the carrier discriminant exponent  $-\frac{5}{4} = q(D_5)$  (seventh digit). The equation is therefore *not* freely tunable to 137.036: the two integers that hit it are exactly the compiler/bootstrap numbers, fixed elsewhere. What it does *not* do is pin the carrier norm strongly — that is honest, and it is the right level of claim. [N]

## Appendix: computational verification

The load-bearing identities of this document (marked [I], [L], [N]) are re-derived from  $\{c_3, g_{\text{car}}\}$  alone and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v1_e8_glue.py` — the  $\mu_4$  glue ( $q(D_5)+q(A_3)=2$ ,  $\text{disc}=\mathbb{Z}_4$ , 240, 248).
- `v2_carrier_pascal.py` —  $g_{\text{car}}=5$ ,  $16=1+5+10$ ,  $N_{\text{fam}}, \Omega_{\text{adm}}, b_1$ .
- `v3_em_alpha.py` — the EM fixed point  $\alpha_\star^{-1}=137.0359992168$  (existence/uniqueness, ablation).
- `v4_flavor_matrix.py` —  $R$  with  $\det=8$ , minors  $\{2, 3, 5\}$ ,  $\sum L=40$ .
- `v7_gravity_cosmo.py` — scalaron  $c_3^7$ ,  $\Lambda/A_s$  readouts.

Run `python verification/run_all.py` (needs `mpmath`, `numpy`, `sympy`); it exits 0 iff all checks pass. Full map: `verification/README.md`.